SERVICE PLANNING PORTFOLIO

Minkyeong Kim

Service Planning Portfolio

A planner who turns customer and market insight into results

※ All data shown in this portfolio are example figures.

Part 01
Enterprise AI Platform
Snowflake → Knowledge graph → MCP / Skills → Agent platform

Part 02
AI Talent Management
Training → AI talent development platform → AI talent evaluation

Part 03
Internal Business Systems Planning
Problem definition → Data structure design → Build → Operate & improve

Part 04
Strategy Projects (Internal Consulting)
Customer and market insight

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10 projects

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01	**Enterprise AI Platform**	AI agent platform (the in-house AI agent) / MD business ontology / PaaS operations
02	**AI Talent Management**	AI agent training / Fashion AI platform
03	**Internal Business Systems Planning**	Image Platform / RAG search
04	**Strategy Projects (Internal Consulting)**	Naver channel strategy / Padded-jacket rebranding / China brand churn analysis

SUMMARY

Summary

Started in customer and market analysis for fashion brands; now building the platform that connects company data to AI agents and designing the company-wide AI transformation. From the numbers to the shipped service, end to end.

No.	Part	Project	Period	Key result
01	**Enterprise AI Platform**	**Knowledge Graph-based Enterprise AI Agent Platform** A knowledge-graph-based enterprise AI agent platform where a plain-language request is enough to build a web app connected straight to the company DW	2025.11 – ongoing	**160** Agents completed
02	**Enterprise AI Platform**	**Building the Ontology: Adding a Business-Concept Layer to the Data Catalog** From a semantic layer centered on DW tables and columns to a business ontology with business-judgment concepts added	2026.11 – ongoing	**markdown rate · reorder quantity · reorder timing · allocation · main sales channel** Wider answer scope (judgment questions)
03	**Enterprise AI Platform**	**PaaS-based Service Operations** An in-house PaaS that deploys the apps non-developers build and provides the DB, file storage and schedules they need	2026.03 – ongoing	**311** Deployments
04	**AI Talent Management**	**Company-wide AI Agent Training** Small, step by step, repeatedly: employees build AI agents hands-on	2026.02 – ongoing	**500 in total** AI agent training attendees
05	**AI Talent Management**	**Fashion AI Platform: Web-App Sharing, Recommendation & Submission Portal** Where employees publish business web apps built with the in-house AI platform, get similar apps recommended by job, and submit results	2026.08 – in operation	**119 / month** Web apps uploaded (monthly average)

No.	Part	Project	Period	Key result
06	**Business Systems**	**Company-wide Integrated Image Platform** Unified management of every image in the company, with a flow built from image registration to delivery to online channels	2025.09 – 2026.03	**6 days → 1 hour** Listing lead time
07	**Business Systems**	**Internal Strategy-Document RAG Search** Not simple search, but a service that searches the organization's decision criteria	2025.01 – 2025.07	**98%** Budget saved
08	**Strategy Projects**	**Redefining the Naver Channel Sales Strategy** Why it was customer-structure redesign, not channel expansion	2024.01 – 2024.02	**137%** Live-commerce conversion growth
09	**Strategy Projects**	**the sports brand Padding Rebranding Strategy** Designing core products from big data	2024.11 – 2024.12	**100%** Sales growth among customers under 30
10	**Strategy Projects**	**Customer Churn Analysis for a Premium Brand in China** Six non-customer studies to find ‘brand casual’, localized for white-collar (白领) women	2025.02 – 2025.04	**+8%** Sales growth vs. same month last year

No. 01 · Part 01 Enterprise AI Platform

Ongoing | Enterprise AI Platform

Knowledge Graph-based Enterprise AI Agent Platform

A knowledge-graph-based enterprise AI agent platform where a plain-language request is enough to build a web app connected straight to the company DW

1 Problem	2 What I did	3 Result
Web apps built by employees could not connect to the company DW, so they saved little real time, and a single BI data query took 27 minutes on average; action was needed.	**Put metric definitions in a knowledge graph so a plain request builds a DW-linked web app.**	**160 agents built and 8,180 work hours saved per month (Sept 2026).**

Building a dashboard by vibe coding with this tool

1 자연어 한 줄로 요청

A브랜드 매장별 매출 보여주는 대시보드 웹 만들어줘

2 대시보드 웹이 바로 생성



A브랜드 매장별 매출 대시보드
localhost:8765, 브라우저에서 열림

열기 ···

3 '연동해줘' 한 마디로 Snowflake 실데이터에 연결

네, 연동해줘

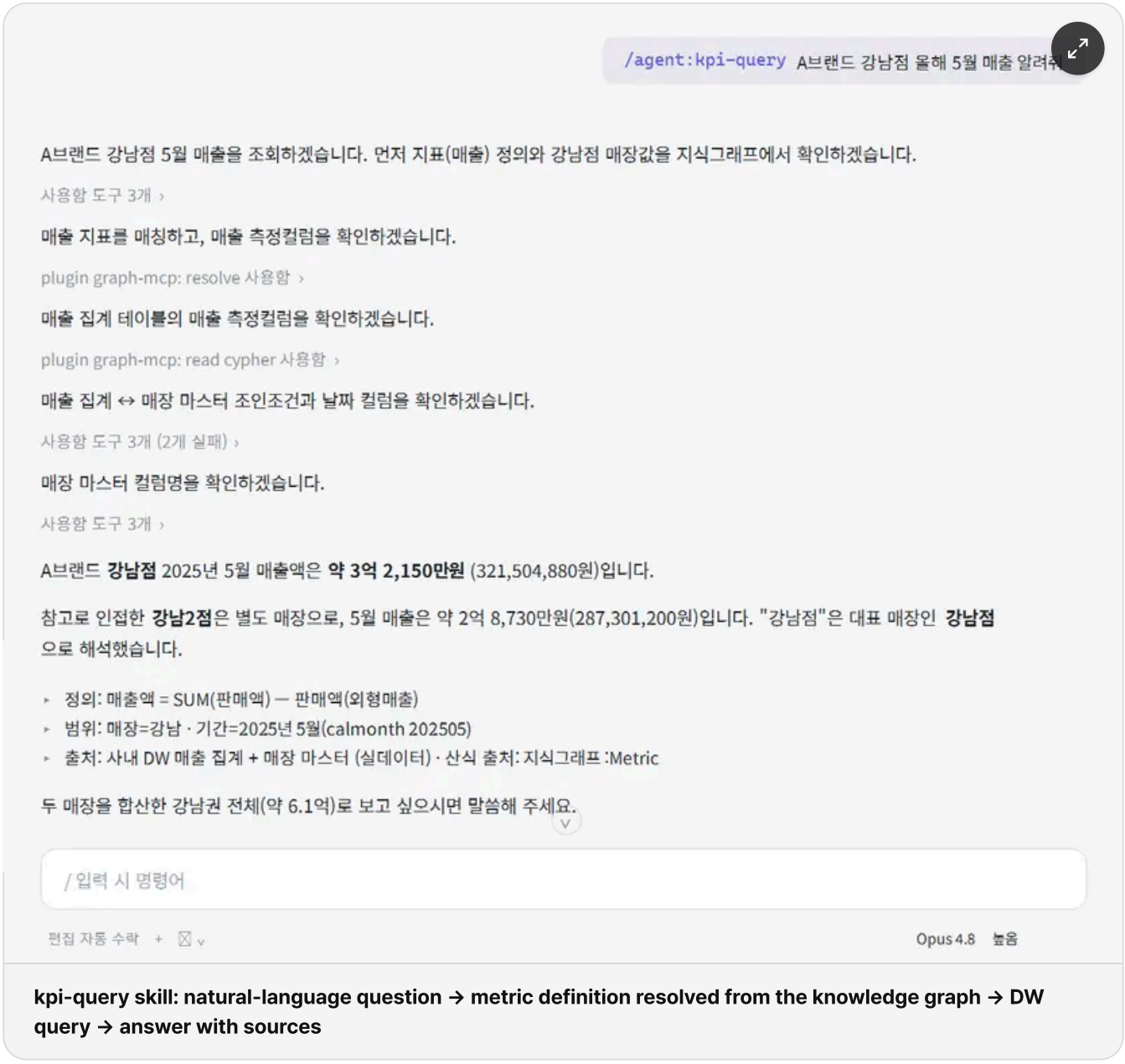
대시보드를 Snowflake 실데이터로 교체했고, 브라우저에서 콘솔 오류 없이 뜨는 것까지 확인했습니다.

데이터 기준

- **출처:** SALES_STORE_DAILY의 판매액(외형매출)입니다. A브랜드는 상품기준(BRAND_CD → PRDT_BRAND)으로 잡았고, 지식그래프 규칙대로 SALE_TYPE='정상', USE_YN='Y' 조건을 걸었습니다.
- **기간:** 2025-01부터 어제까지 월별로 집계했고, 데이터는 data.js에 저장했습니다.
- **매장 수:** 34개입니다. 물류센터·BU 등 채널 정보가 비어 있는 코드 12개는 매장이 아니라고 보고 뺐습니다.
- **현재 수치:** 2026년 총매출은 1,024.3억원입니다. 1위는 온라인스토어(132.7억원)입니다.

Vibe-coding demo: one word, 'connect it', links it to live Snowflake data

Querying BI data in plain language



kpi-query skill: natural-language question → metric definition resolved from the knowledge graph → DW query → answer with sources

Purpose PURPOSE

1. Build web apps connected to the company DW with plain-language requests alone

2. Query BI data in plain language

Results RESULT

160

Agents completed

as of Sept 2026 (first month after launch)

8,180 h / month

Work hours saved by agents

as of Sept 2026 (first month after launch)

85% less

Time to query in-house (BI) data

AS-IS 27 min → TO-BE 4 min

Background BACKGROUND

1. Web apps built by employees could not connect to the company DW, so they saved little real time; the aim was to connect individually built web apps to in-house data while keeping security issues to a minimum.
2. Querying in-house BI data took 27 minutes on average; the aim was to cut that time by 75% with AI.

Project contents CONTENTS

System architecture

This platform puts agents on top of a knowledge graph. Built on a Neo4j knowledge graph, it processes each question in the order organization instructions → hooks → skills → MCP tools → gateway → knowledge graph and internal data.

1. Client Layer

The company plugin in Claude Desktop, Claude Code or Cowork

Claude Desktop · Code · Cowork

company plugin
bundle of skills, hooks and MCP connectors

Org connector registration
deployed per organization by admins

SSO login
queries run with the user's own permissions

2. Policy & Context Layer

Org instructions and hooks read the context graph

Org instructions
rules for everyone, e.g. what to do when an MCP connection fails

Session-start hook
personalization, memory and in-house PaaS development rules

Pre-question hook
adds past corrections, golden SQL and company decisions from the context graph

Safety principles
read-only · own permissions · no personal-data columns · verify the person in charge is still employed

3. Agent Runtime

The Claude model picks a skill and calls tools in order

Claude model
reasoning · SQL writing · answer writing

Skill triggers
skills selected by keyword and request type

Memory
per-user preferences and team rules kept as files; new definitions are recommended for knowledge graph registration

4. Capability Layer (skills)

Skills hold no rules; they only set how to query the graph and which tools to call

Natural-language KPI skill
one graph lookup → SQL → read-only DW run

Weekly report search skill
finds brand weekly reports via the graph's vector index

Internal admin guide skill
expenses, policies, general affairs and other internal knowledge search

App build & deploy skill
building and preparing apps for deployment on the company standard

5. Tool Binding Layer (MCP tools)

The tools skills actually call

Knowledge graph lookup tool
metric meaning · graph reads · weekly report and admin search

Company DW query tool
runs Snowflake queries (read-only)

HR (attendance) tool
leave info, requests, approval lines, approve, cancel

Mail & calendar tool
read, send and reply to mail; create and update events

6. Knowledge Graph & Data Layer

Through the in-house MCP gateway, the DW, M365 and HR are read by the knowledge graph's definitions

Neo4j knowledge graph
metric definitions, formulas, joins, golden SQL

Snowflake DW
standard data models (aggregate tables)

M365 Graph
mail and calendar

the in-house PaaS
one-click deployment of employee-built apps

How a question is processed

1

Question

Ask in everyday words, e.g. “What were Brand A Gangnam's sales this May?”

2

Knowledge graph lookup

One graph lookup returns the metric definition and joins; SQL is written only from them

3

Answer with sources

Read-only DW query with the user's own permissions; definition, scope and sources attached

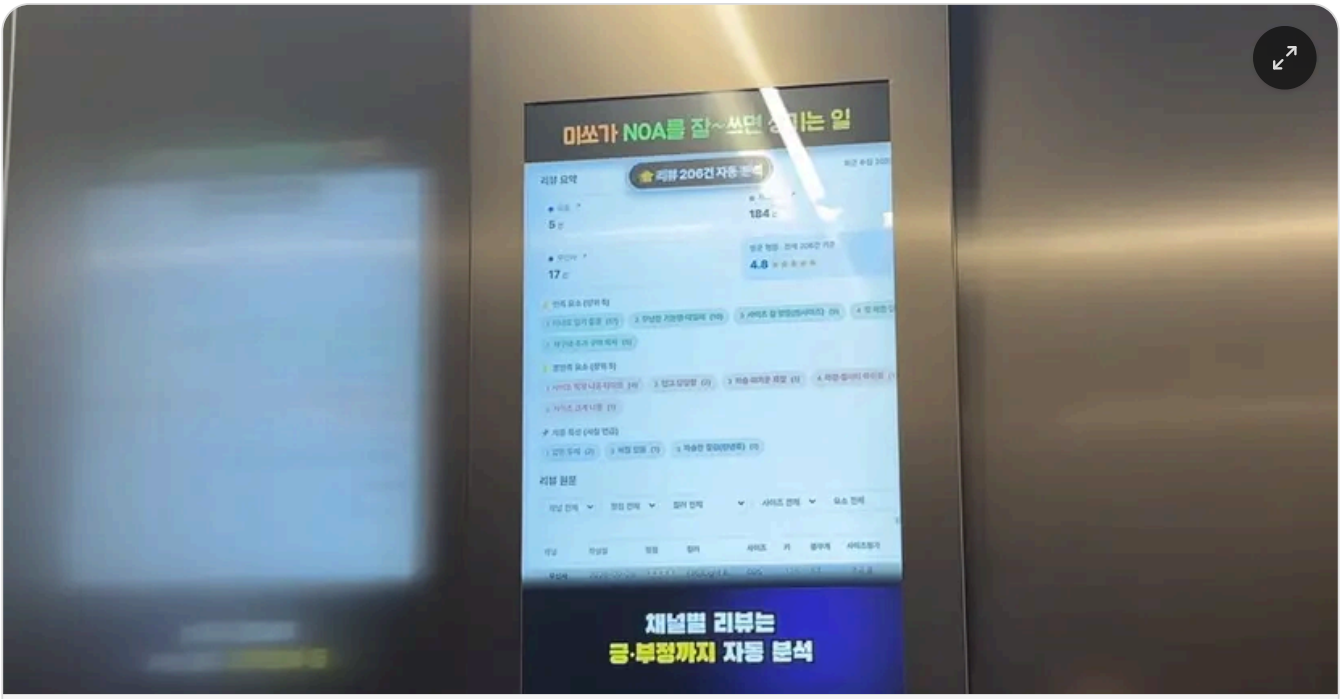
4

Learning from corrections

A correction like “No, from 25 January” is saved to the context graph and applied from the next question

Process

1	2	3	4	5
Knowledge graph design and build	**MCP servers connecting the graph and the DW**	**Skills and hooks**	**Plugin rollout and offline training**	**Weekly use-case broadcasts across affiliates**



Step 5 in practice: a weekly use-case video broadcast on office elevator screens (automatic positive/negative review analysis by channel)

Role & contribution ROLE

1. Built a mapping dictionary between business terms (brand, season, store) and DW tables and columns: organized synonyms and reference values so the same question gets the same answer
2. Defined requirements for the MCP server linking the graph and the DW: agreed the tool list, input/output formats and read-only permission scope with developers
3. Designed the plugin rollout process: wrote the install guide, connector setup steps and a troubleshooting guide for common errors

※ All data shown in this portfolio are example figures.

ongoing Enterprise AI Platform

Building the Ontology: Adding a Business-Concept Layer to the Data Catalog

From a semantic layer centered on DW tables and columns to a business ontology with business-judgment concepts added

1 Problem

Company judgment rules weren't in the data catalog, so users re-entered them each session.

2 What I did

Drew out five weekly MD decisions in interviews and designed an ontology layer.

3 Result

AI now answers judgment questions such as markdown rate and reorder quantity and timing.

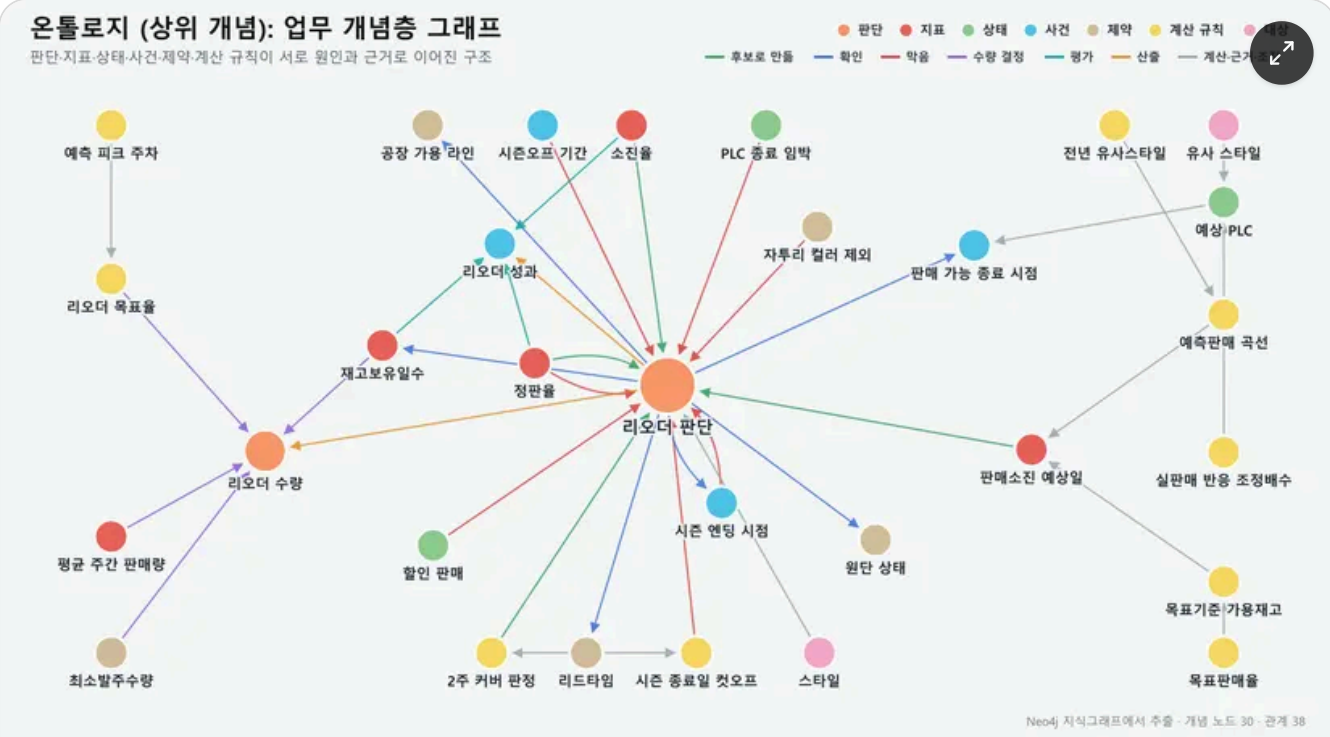
Glossary

Full-price sell-through · Share of stock sold at full price, without discounts

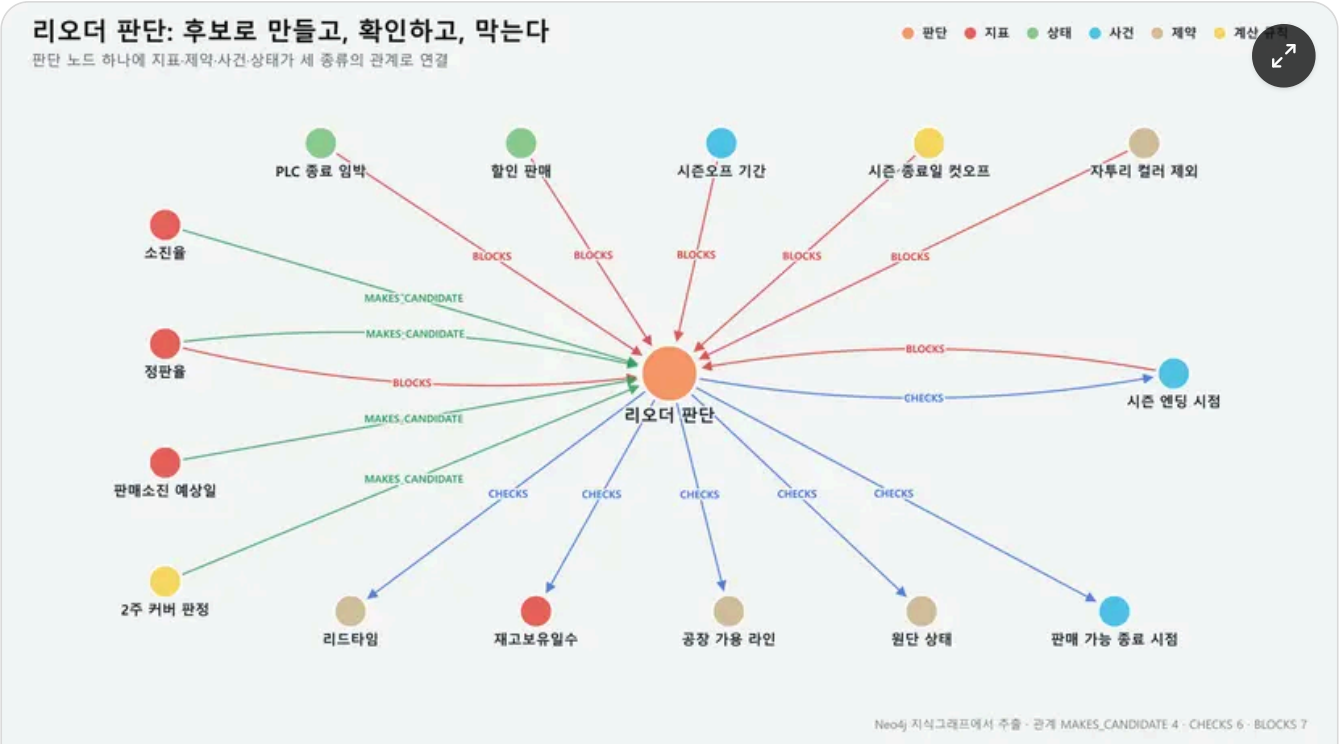
Sell-out rate · Share of received stock that has been sold

PLC · Product life cycle: introduction, growth, peak, decline

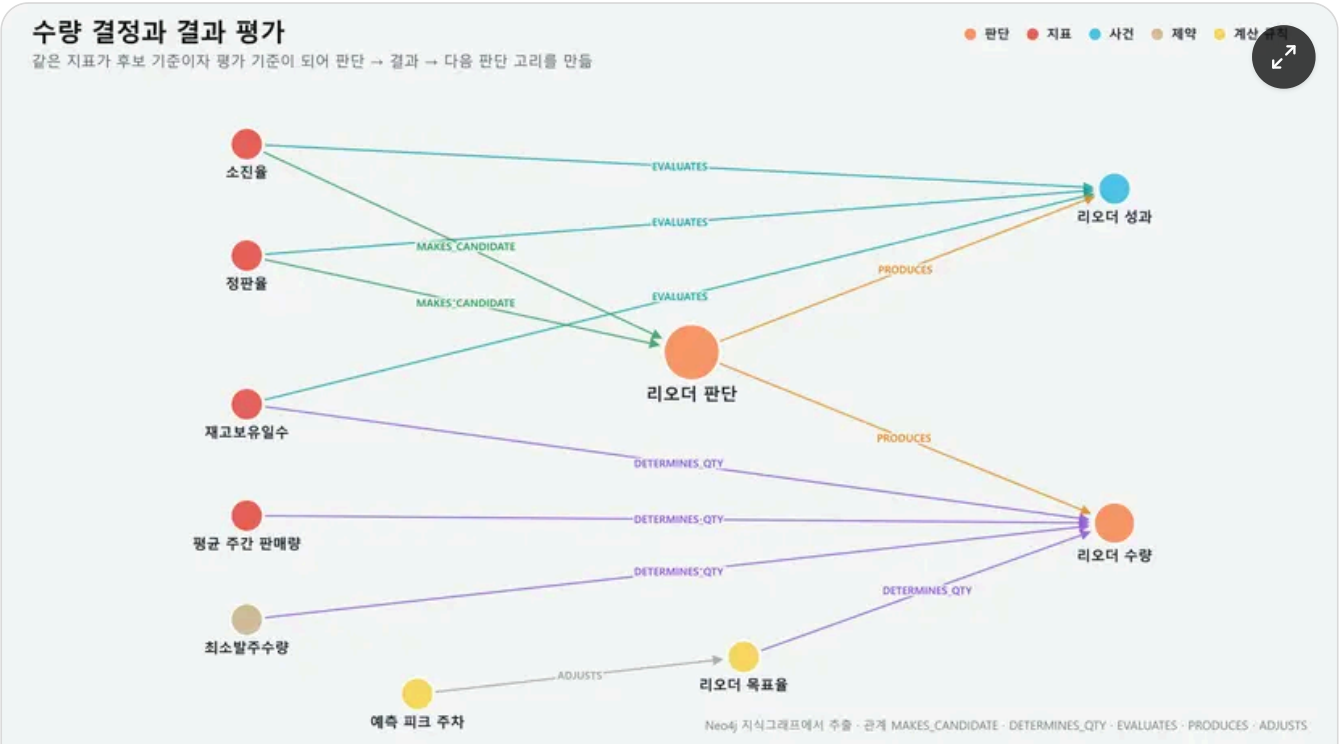
Reorder · Ordering more of a product that is selling well



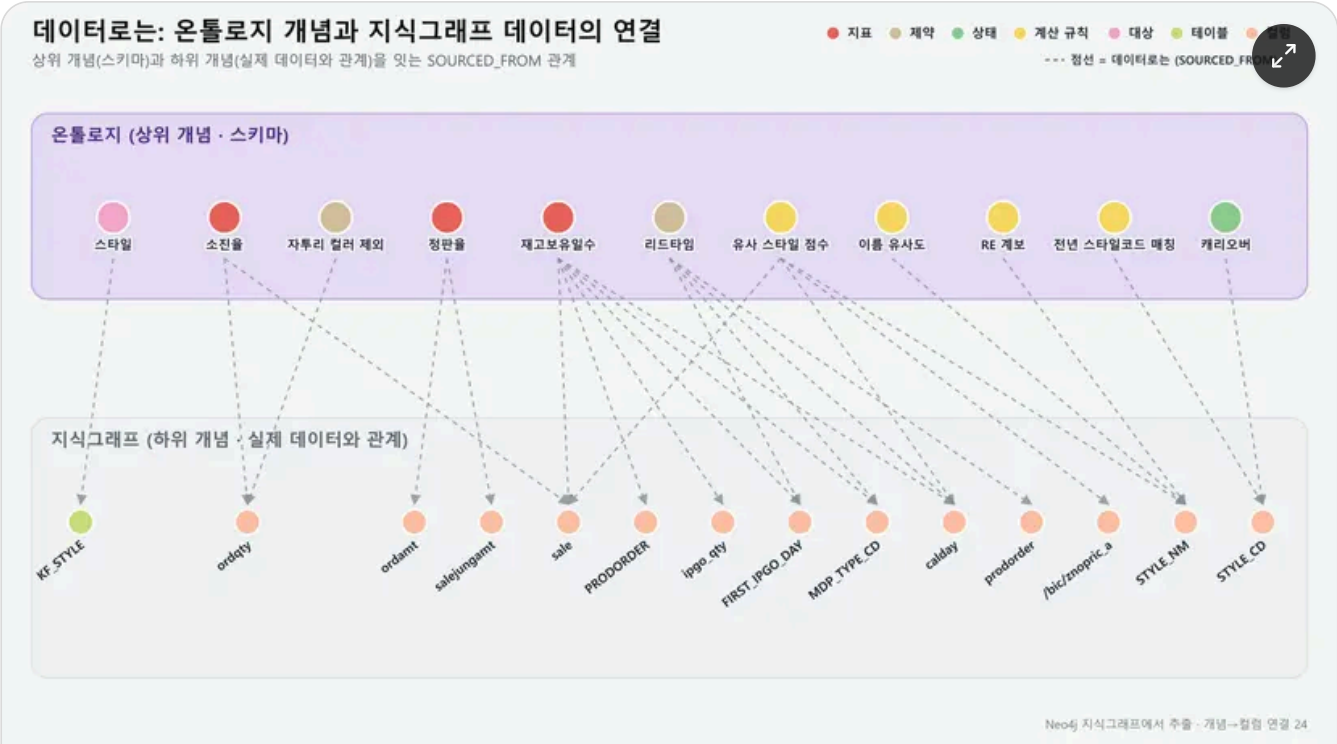
Ontology (upper concepts): the business-concept layer where decisions, metrics, states, events, restrictions and calculation rules are linked as causes and grounds (30 nodes · 38 relations)



Reorder decision: metrics, restrictions, events and states make it a candidate (MAKES_CANDIDATE), are checked (CHECKS) and block it (BLOCKS)



Quantity and evaluation: the same metrics serve as candidate criteria and evaluation criteria, forming a decision → result → next decision loop



Seen in data as: SOURCED_FROM relations tie upper concepts (ontology) to lower concepts (knowledge graph) (24 links)

Period	2026.11 – ongoing
Area	Ontology · MD knowledge · Knowledge graph
Stack & tools	Neo4j · Cypher queries · Snowflake

Purpose PURPOSE

Extends the range of answers to judgment questions by building an ontology that holds business know-how on top of the existing data catalog (semantic layer)

Results RESULT

markdown rate · reorder quantity · reorder timing · allocation · main sales channel

Wider answer scope (judgment questions)

Background BACKGROUND

The company's unwritten rules and ways of working were not in it, so even with a semantic layer, users had to enter the company's judgment criteria again in every session.

Project contents CONTENTS

System architecture

The existing knowledge graph (semantic layer) serves as the DW data-link layer, while MD experience knowledge such as 'fit complaints → size issue → more returns → lower full-price sell-through' is added to support faster decisions

1. Ontology (upper concepts · schema)

Nouns become concepts, verbs become relations. Concepts are grouped into buckets (labels) from the MD world, not the DW.

Metrics
sell-out rate · full-price sell-through · days of stock · early sell-through

States / stages
PLC early · growth · peak · decline

Decisions
reorder · channel focus · markdown timing · next-season orders

Constraints
days of stock · lead time

Timing / events
season end · off-season sale · forecast peak passed

2. Knowledge graph (lower concepts · real data and relations)

Says where each concept is visible in data. Nothing built so far is thrown away; it keeps the upper layer from misreading data.

Full-price sell-through → DW sales metrics

Fit complaints → review text · platform detail ratings

Guarding facts
the repair/inspection CS channel is not customer reviews · company-mall photos are not collected · jewelry accumulates lines per receipt

How a question is processed

1

User judgment query

“Why isn't this style selling? What should we do?”

2

Find the decision node

Check which PLC stage it is in, such as introduction or decline.

3

Seen in data as

Confirm exact numbers through the lower-level knowledge graph: full-price sell-through from sales metrics, fit complaints from review-text tables.

Process

1	2	3	4
Interview SO/MD staff (judgment criteria in the work)	**Write concept and relation tables**	**Migrate to Neo4j**	**Connect the Neo4j MCP**

Role & contribution ROLE

1. Organized business concepts and causal relations through interviews with practitioners
2. Defined the concept-table and relation-table formats (direction, attributes, exceptions)

※ All data shown in this portfolio are example figures.

Ongoing Enterprise AI Platform

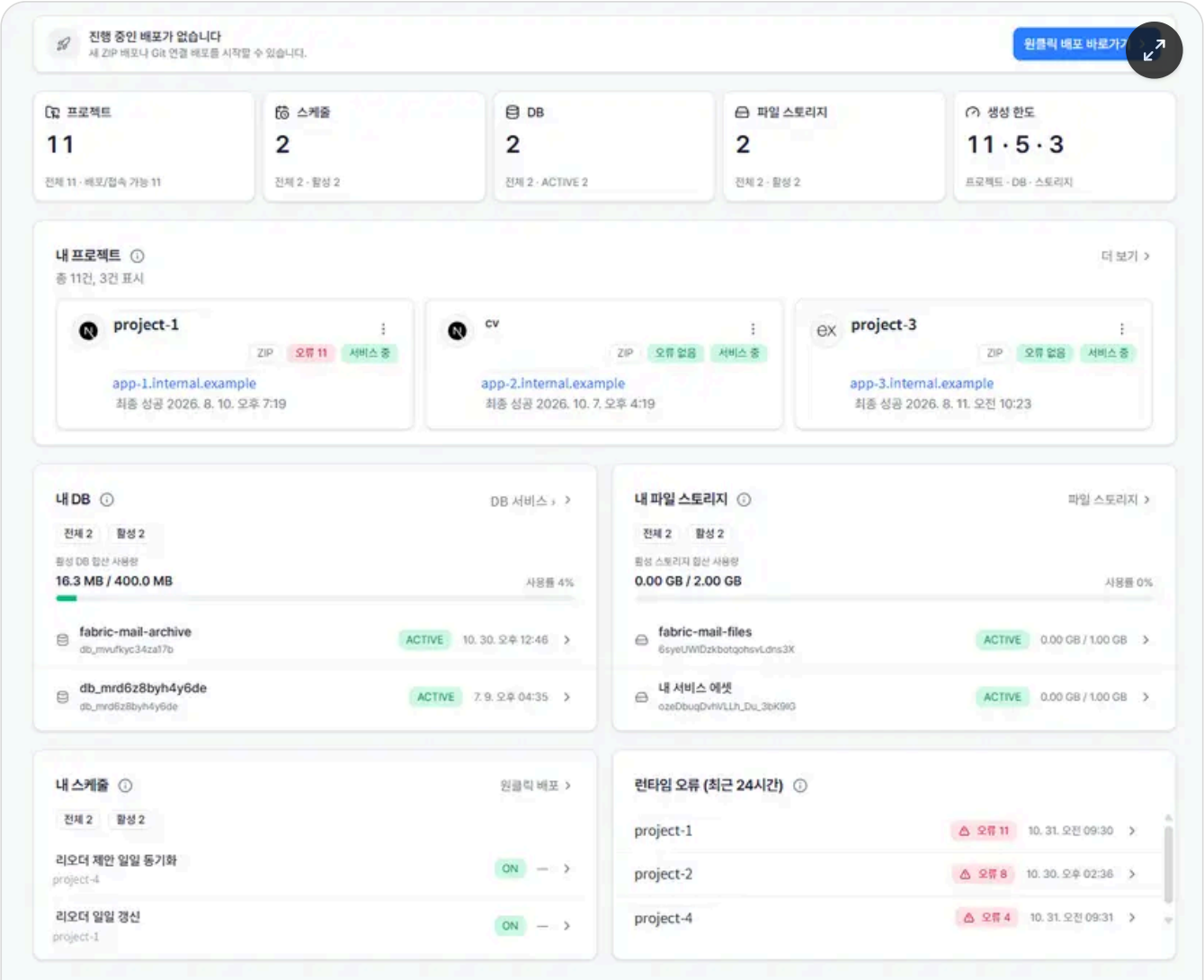
PaaS-based Service Operations

An in-house PaaS that deploys the apps non-developers build and provides the DB, file storage and schedules they need

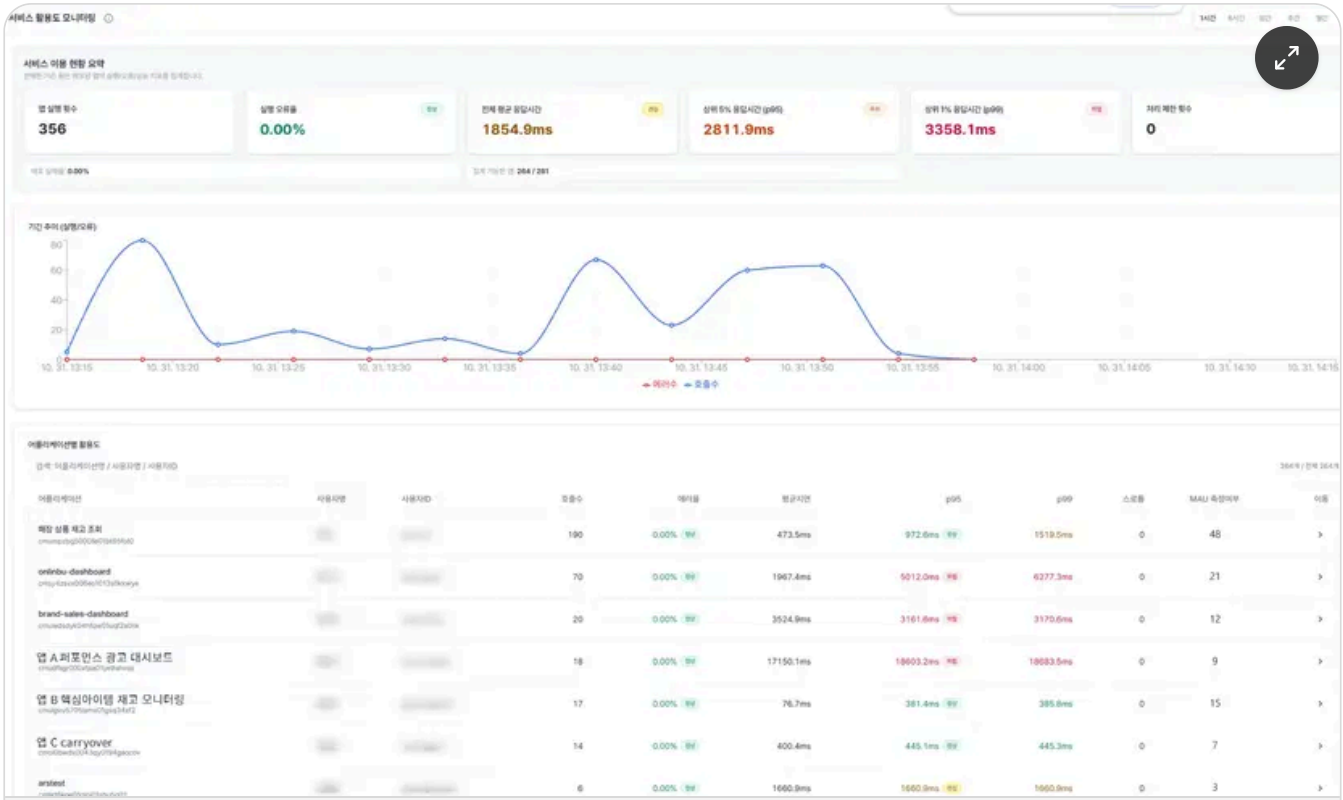
1 Problem	2 What I did	3 Result
Employee-built web apps sat on separate servers and outside DBs, hiding data outflow.	**Built an in-house PaaS for deployment, DBs and schedules, plus guardrails for AI tools.**	**311 deployments with a 90% deployment success rate (n=235).**

Glossary

MAU · Monthly active users: people who used an app in a month



The status of the web apps each person has deployed, as seen on their own dashboard



Admin view of every deployed web app

Period	2026.03 – ongoing
Area	PaaS · One-click deployment · Self-service
Stack & tools	Next.js · Serverless · CloudFront · Claude Code Skill

Purpose PURPOSE

An in-house PaaS where line staff deploy the web tools they build without developer help and get operations support. Overall deployment status, per-app MAU and call counts, and deployment domains are managed in one place.

Results RESULT

311

Deployments

90%

Deployment success rate

n=235 deployments

Background BACKGROUND

Web services built by individual employees were deployed to individual servers, which made company information hard to manage and made it hard to see which company data left through external DBs and external hosting.
The in-house PaaS was built to bring deployment inside the company, manage it in one place, and consolidate internal information, security and cost.

Core features CORE FEATURES

Core feature 01 One-click deployment

Upload a ZIP or connect a Git repository and the app deploys. Guardrail rules for AI coding tools were shipped first to raise the success rate.

ZIP and Git deployment

Deploy by uploading a finished ZIP or connecting a Git repository. Each project card shows deployments in progress, the last successful time and error counts.

Deployment guardrails

Rules that AI coding tools such as Cursor and Claude Code follow when generating code (platform and stack choices, safe defaults), distributed as a document to cut failed deployments.

Pre-deployment skill

A Claude Code skill checks project structure, environment variables and build settings against PaaS rules before deploying.

Core feature 02 Operations, security and domains on one screen

Usage of each deployed app (MAU and call counts), data-security checks, deployment domains, schedules, DBs and file storage are all managed from one dashboard.

MAU and call counts per app

Monthly active users and API call counts are shown for every deployed app, separating the apps that are really used from the ones to retire and putting operating results in numbers.

Data-security check

Company data must stay in the internal DB, so each app is checked before deployment for connections to any external database, closing the path for company data to leave.

Deployment domain management

Domains are issued, changed and revoked per app, and which domain belongs to which app and owner is managed in one place.

Projects, schedules, DB and file storage

Deployed apps with their service status, jobs that run at set times, and the databases and file stores apps use are created and monitored here.

Creation limits

Project, DB and storage limits are shown so resources are planned.

Role & contribution ROLE

1. Wrote, distributed and taught the one-click deployment guardrails (rules for AI coding tools)
2. Designed Claude Code skills (requirements, triggers, behavior) that check a project against PaaS rules before deployment
3. Turned recurring deployment and operations inquiries into skill features

※ All data shown in this portfolio are example figures.

Ongoing AI Talent Management

Company-wide AI Agent Training

Small, step by step, repeatedly: employees build AI agents hands-on

1 Problem

The goal was 20% AX adoption, but most employees didn't know where to start.

2 What I did

Designed small, level-based, repeated classes and a 6-step agent-building practice.

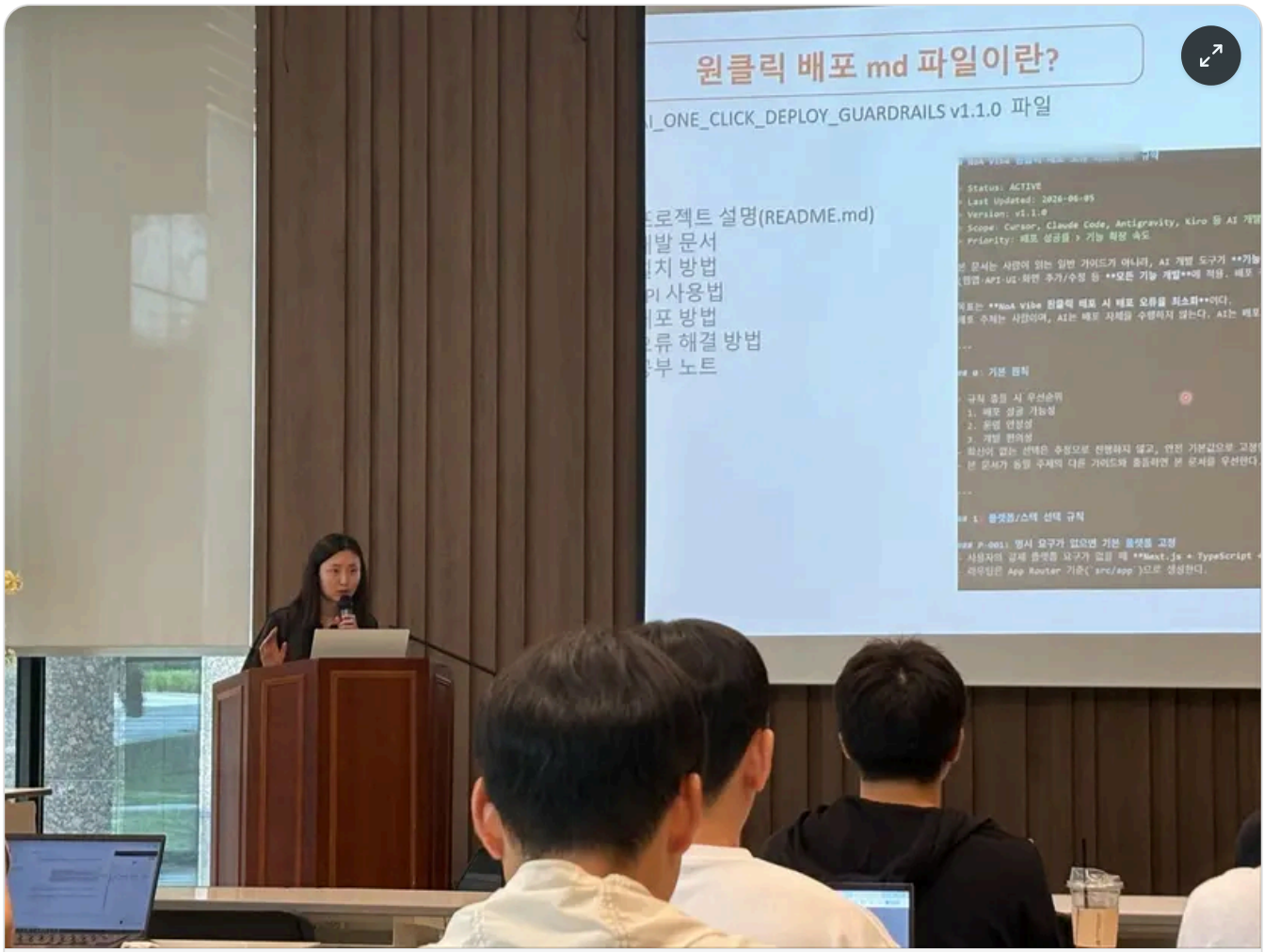
3 Result

500 people trained in total, each finishing one web tool of their own per session.

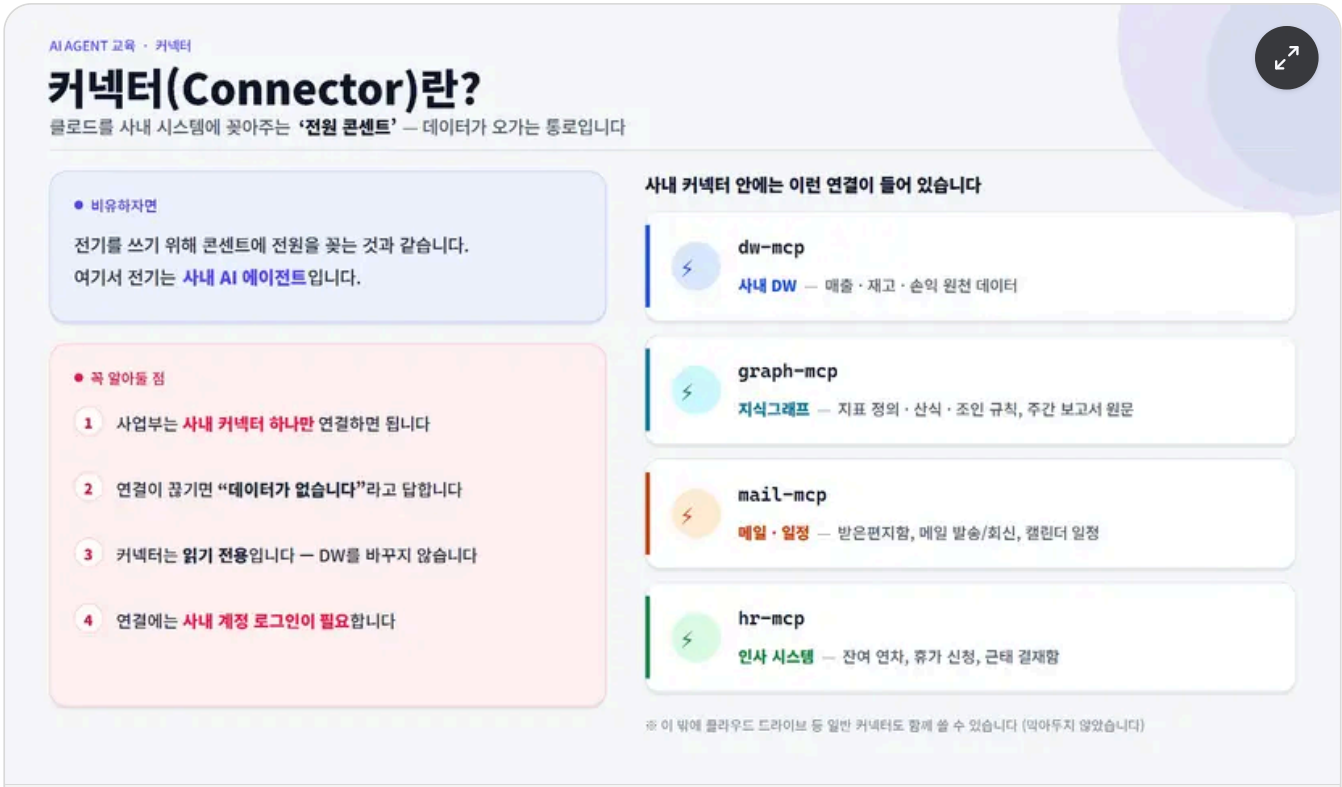
Glossary

AX · AI transformation: changing how work is done with AI

Hands-on session · A practice class where people build things themselves



AI agent-building training: one-click deployment guide session



Connector training material

Period	2026.02 – ongoing
Area	AI agent training · Training · Onboarding

Purpose PURPOSE

Designed training, onboarding and operational support to reach a 20% AX conversion rate across the organization.

Results RESULT

500 in total

AI agent training attendees

1 per session

Each attendee finishes one web-based tool of their own by the end of every session

Background BACKGROUND

To reach 20% AX adoption in-house, small-group, level-based offline training is repeated for employees who do not know where to start.

Project contents CONTENTS

Process

1	2
Hands-on agent training	**Build & deploy support**

Training content TRAINING CONTENT

Part 1 Level-based concept training

Because skill levels differ widely, one lecture is too hard for beginners and dull for experienced staff, so classes are split by level and kept small. Concept training is the preparation for the hands-on session that follows.

AI literacy basics

Fourteen slides on AI fundamentals and what matters at work. Security principles and one-click deployment guardrails rewritten in plain company language.

Classes by level

Beginner, intermediate and advanced classes teach concepts at the right level, removing the problem of a single lecture being hard for beginners and dull for experienced staff.

in-house AI agent 2.0 training decks

Classroom decks in two sessions: session 1 (concepts, 41 slides) and session 2 (hands-on, 30 slides).

Connector concept

Explains connectors with a power-outlet analogy instead of developer terms, covering four things

Part 2 Hands-on sessions

Right after concept training, even first-time attendees turn a problem from their own work into one agent and publish it. Completion means one web-based tool of their own by the end of every session.

6-step agent-building practice

Understand AI → define the work problem → write a prompt spec → prototype with Claude Code → security check → prepare to share and deploy. Each step has an output and check questions, with first-agent scenarios for MD, sales, design and CS.

5-step vibe-coding guide

Decide what to build → build screens by vibe coding → connect BI data → prepare code for deployment → deploy. A usage-guide prompt makes the builder state operating logic, team tacit rules and KPIs first.

One-click deployment

The finished tool is deployed to the in-house PaaS in one click, with a deployment guide session covering guardrail rules and the deployment procedure.

Build and deploy support after the session

FAQ, 1:1 inquiries, data requests and offline help sign-up resolve whatever blocks people after the session.

Role & contribution ROLE

1. Designed the curriculum and job-specific hands-on exercises

2. Built the training content: level-based concept training and hands-on sessions (building a web dashboard)

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Ongoing AI Talent Management

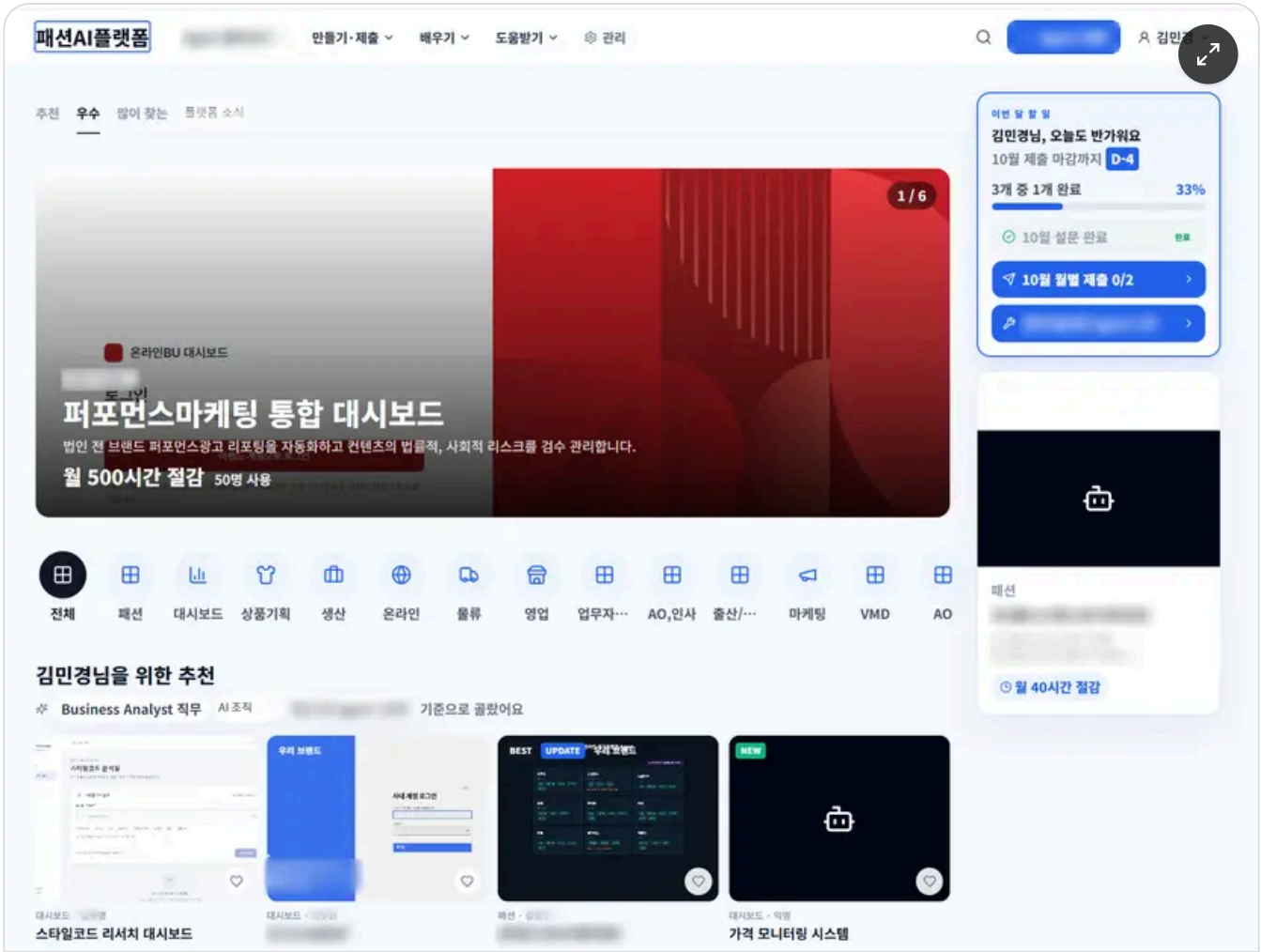
Fashion AI Platform: Web-App Sharing, Recommendation & Submission Portal

Where employees publish business web apps built with the in-house AI platform, get similar apps recommended by job, and submit results

1 Problem	2 What I did	3 Result
Business web apps built by employees, and their results, were scattered.	**Planned an in-house portal and user flow for web-app registration and monthly results.**	**119 web apps uploaded per month on average, with AX data building up automatically.**

Glossary

AX · AI transformation: changing how work is done with AI



Fashion AI Platform home: best web-app banner · categories · monthly to-do sidebar

Period	2026.08 – in operation
Area	Registration & submission · Next.js
Stack & tools	-

Purpose PURPOSE

Employees upload the business web apps they built with the in-house AI platform and browse apps from the same job, and check their own AI capability level

Results RESULT

119 / month

Web apps uploaded (monthly average)

since 2026.09

Background BACKGROUND

Business web apps built with the in-house AI platform multiplied, but who built what and what it achieved was scattered.

Register the web app → submit monthly results → evaluate → train now runs as one flow that records AX status automatically.

Core features CORE FEATURES

Core feature 01

Home: tasks and recommended reference agents the moment you arrive

The first screen shows this month's tasks together with web apps built by colleagues in the same job that fit me

Best web-app banner

Web apps with the BEST badge rotate in a six-slide banner with monthly hours saved and users

Job and category icon filter

One row of job icons (Fashion, Dashboard, Merchandising, Production…); tap to filter web apps

This month's tasks sidebar

Deadline D-day, progress and survey status pinned on the right of every screen

Core feature 02

Web-app registration and monthly submission

A form that states what changed in numbers, plus monthly results; the source data for evaluation and the talent pool

4-part form

① Description ② Usage guide ③ What changed (hours saved, KPI) ④ Access. Missing fields listed with reasons

Usage-guide prompt: eight items that capture know-how

Paste the prompt into the AI for a 1,500+ character guide; operating logic and work tips matter most

Core feature 03 Requests and inquiries that accumulate automatically

Missing data or stuck on usage? Request or ask in the platform; recurring questions become FAQ

Data requests

Request what data is needed and why; tracked by status and turned into DW and graph tasks

FAQ that builds itself

Answers to recurring questions go to the FAQ on home and Help, so repeats fall

Admin morning digest email

Every day at 8 a.m., new accounts, submissions, uploads and survey responses emailed to admins

Role & contribution ROLE

1. Service planning and information architecture
2. User flow design: UI/UX and menus such as issuing a PAT, so people use the site without friction
3. Monthly submission flow design

※ All data shown in this portfolio are example figures.

Done Business Systems

Company-wide Integrated Image Platform

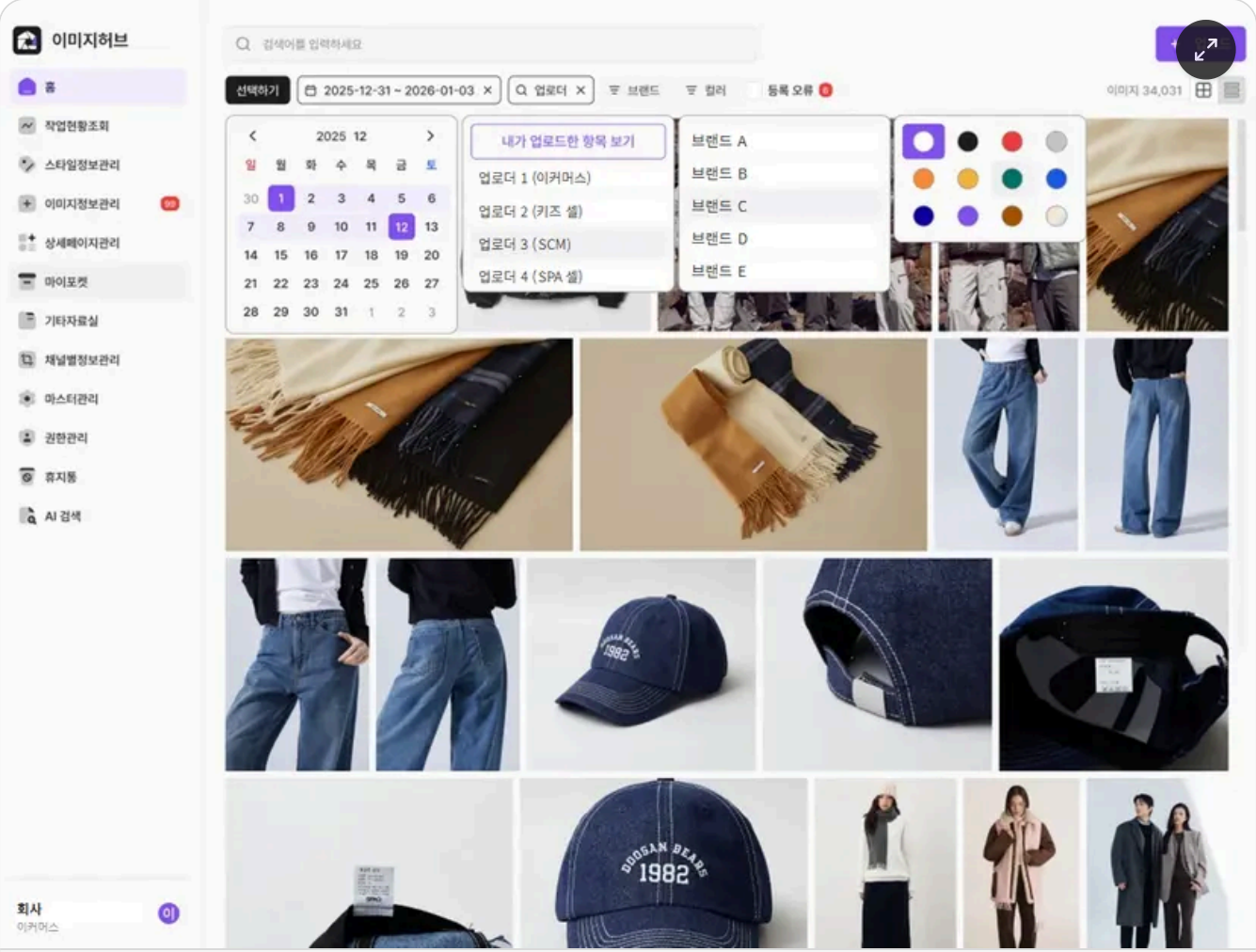
Unified management of every image in the company, with a flow built from image registration to delivery to online channels

1 Problem	2 What I did	3 Result
Scattered product data was re-keyed by hand per channel: 54% of repetitive work.	**Designed a flow where a company-wide integrated image platform and AI format listings and send them to every channel.**	**Listing lead time cut from 6 days to 1 hour; standard listings 100% automated.**

Glossary

Lead time · Time from receiving stock to going on sale

Style code · A unique ID number given to each product



Integrated image platform home: upload date, uploader, brand and color filters plus a ‘registration error’ badge locate any of tens of thousands of images in the grid

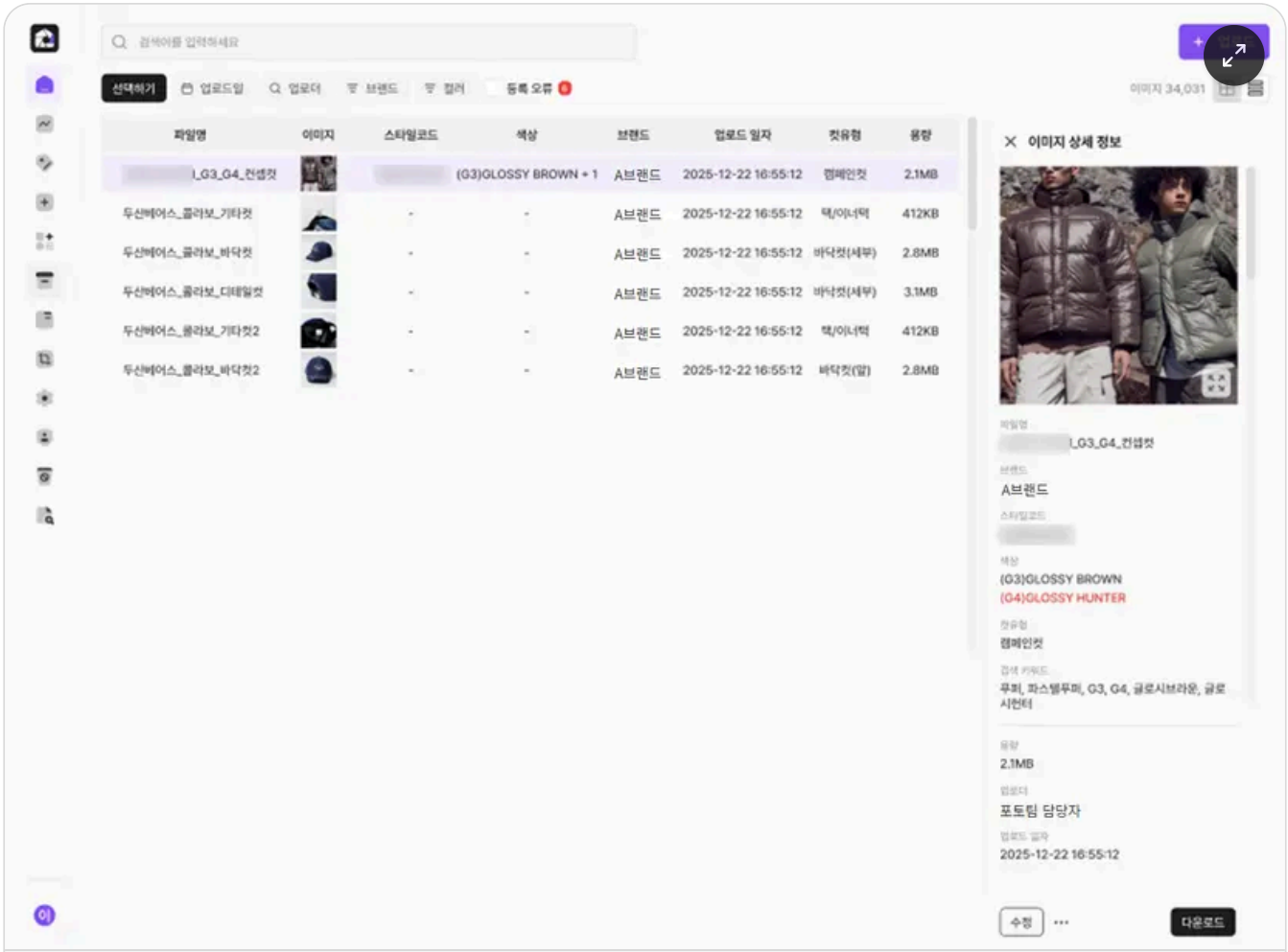
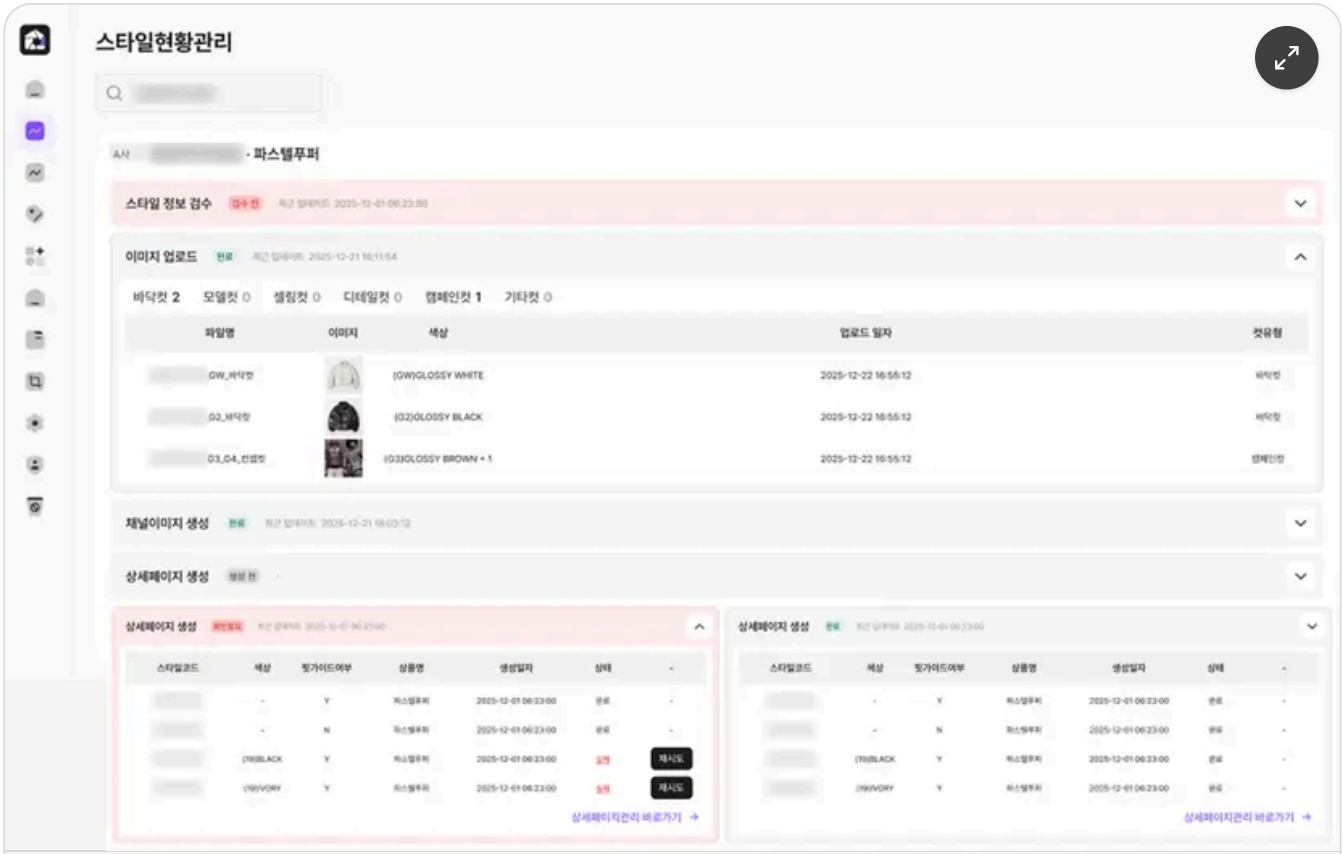


Image registration review and detail check



Style status: image upload → automatic per-channel image editing → review of the detail-page generation stage

Period	2025.09 – 2026.03
Area	Integrated image platform · Product information · AI channel listing
Stack & tools	-

Purpose PURPOSE

Manage every product image in the company (model shots, flat-lay shots, thumbnails) in one company-wide integrated image platform, and let AI adapt registered images and product data to each channel's specs and send them straight to online channels. The goal is to end re-entering the same information channel by channel.

Results RESULT

6 days → 1 hour

100%

Listing lead time

from receiving to on sale

Standard listings automated

only events and promotions remain: 80% of time saved

Background BACKGROUND

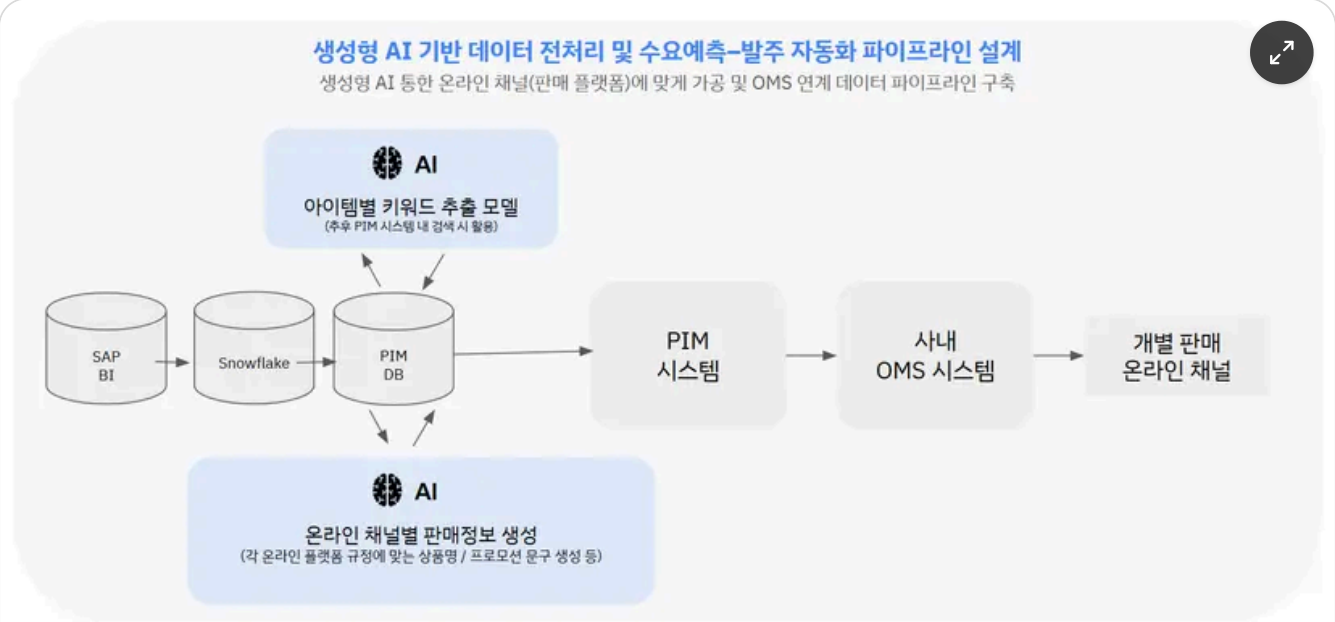
Interviewing MD and AMD staff for three weeks and surveying their work showed that 54% of simple, repetitive work was ‘product listing and management’ (1,865 hours per month). The core bugs were ① product data scattered across SAP, BI, local folders, drives and email, and ② manual adjustment to each channel's different specs.

The project set out to solve this.

Project contents CONTENTS

System architecture

Listing after receiving runs as one flow: images enter the platform, AI builds detail pages, per-channel agents fit each format and every channel receives the listing. Below: the integrated image platform's six layers.



Integrated image platform pipeline: product data in → AI processing (keywords, channel listings) → OMS → online channels

1. Bulk upload

Drag-and-drop bulk upload: re-uploads notify the detail-page AI

Drag-and-drop bulk upload
style code, color and cut type recognized from the file name

Re-upload
added model shots etc. → the page-generation AI notified

Error badge
unmatched files flagged on the home screen

2. Image management

Type, traits and product data turn each image into a searchable asset.

① Image type
front/back/detail, product/model/concept shot

② Product traits
material · mood

③ Style-code link
SAP data (name, price) joined with image platform disclosure data

3. Auto-editing

Images are adjusted automatically to channel specs.

① Resizing
size · ratio

② Retouching
brand logo removal, background removal and more

4. Search

Search by product data, similar image or plain language.

Product-information search
style code · item name · color

Similar-image search
find images with an image

AI search
natural-language search for product images

5. Download

Select the images you need and download them in bulk.

Select and bulk download

My Pocket
keep frequently used images

6. Share

Bundle only the images you want to share and edit.

Shared image bundles

Edit inside a shared bundle

Product listing flow

1

PLM product registration

Base data registered in PLM flows into the image platform automatically.

2

Receiving and inventory → new-product alert

Receiving triggers an automatic new-product alert to the listing team.

3

Shooting · 3D generation → integrated image platform upload / API

Shots (later 3D renders) go straight to the integrated image platform; no more Google upload or email.

4

Detail page production (AI)

AI generates models, sales copy, detail pages and SEO tags.

5

Thumbnail and detail-page upload

Thumbnails and detail pages attach to the style code automatically.

6

Per-channel data-optimization agent

An agent converts names, categories and image specs for each channel.

7

the listing hub listing → all channels

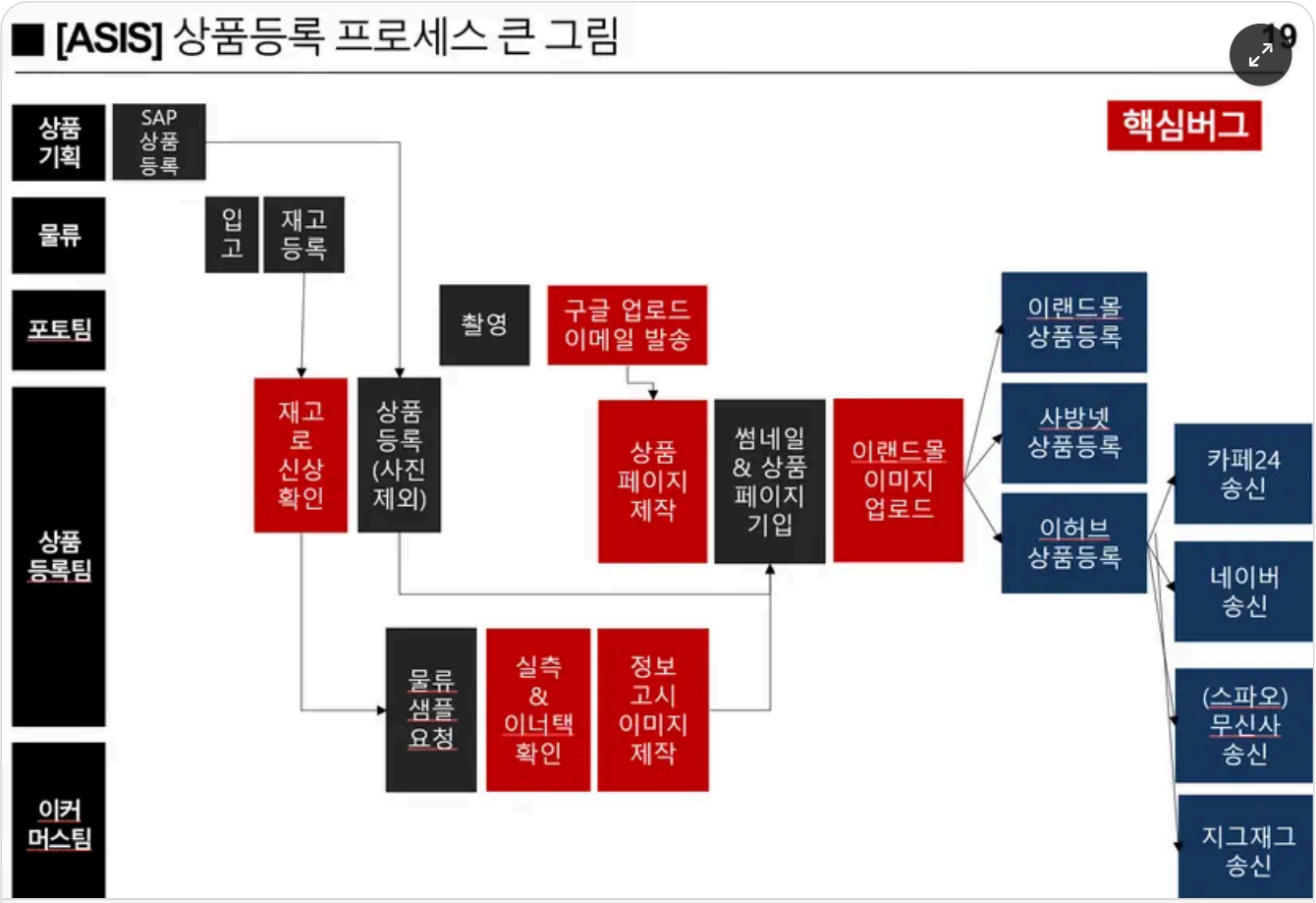
One listing goes to every channel: the company mall, Naver, Musinsa and more.

8

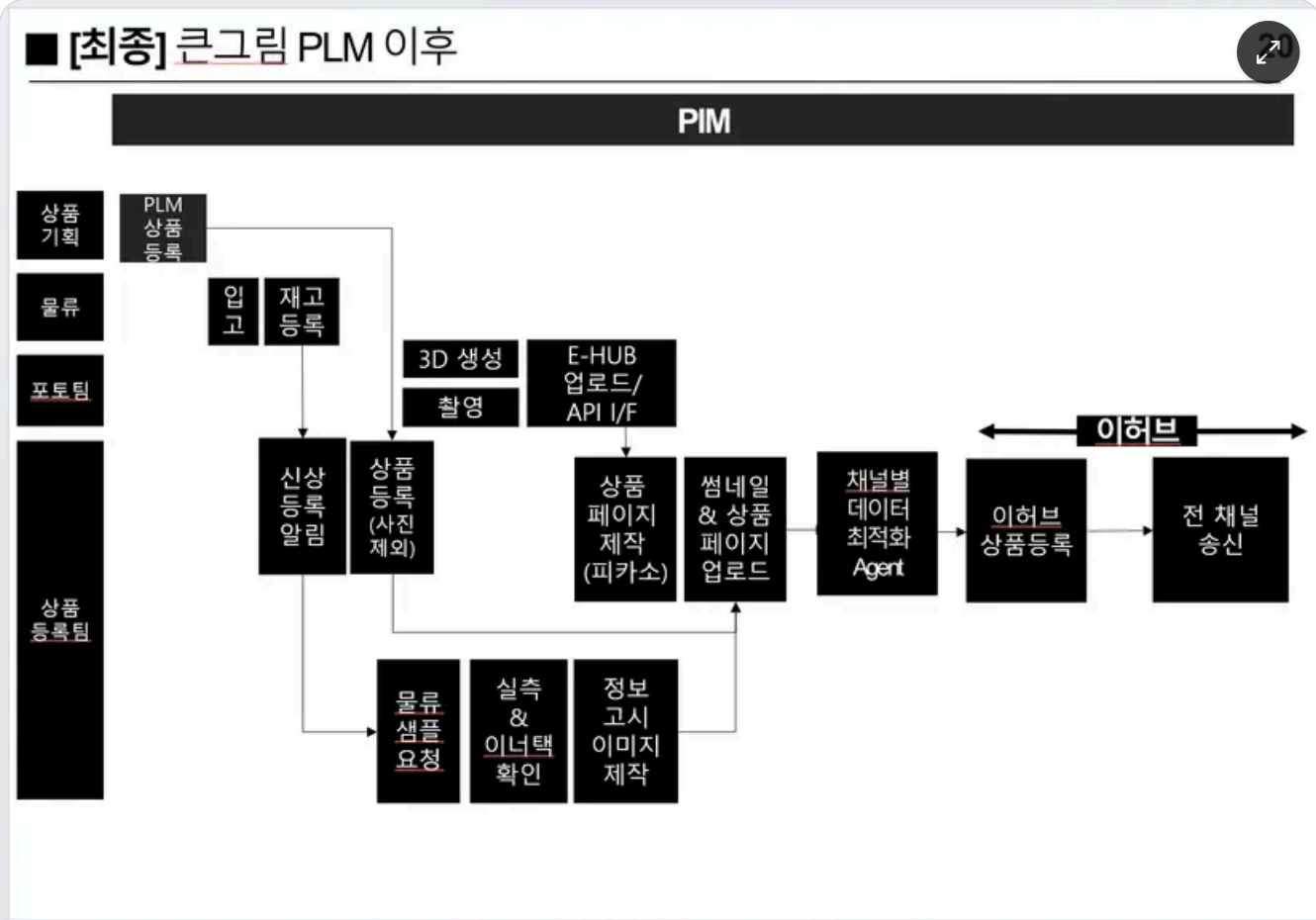
Monitoring dashboard

Blocks ‘received but never on sale’ errors; people only review.

Business process: AS-IS → TO-BE, and the five changes

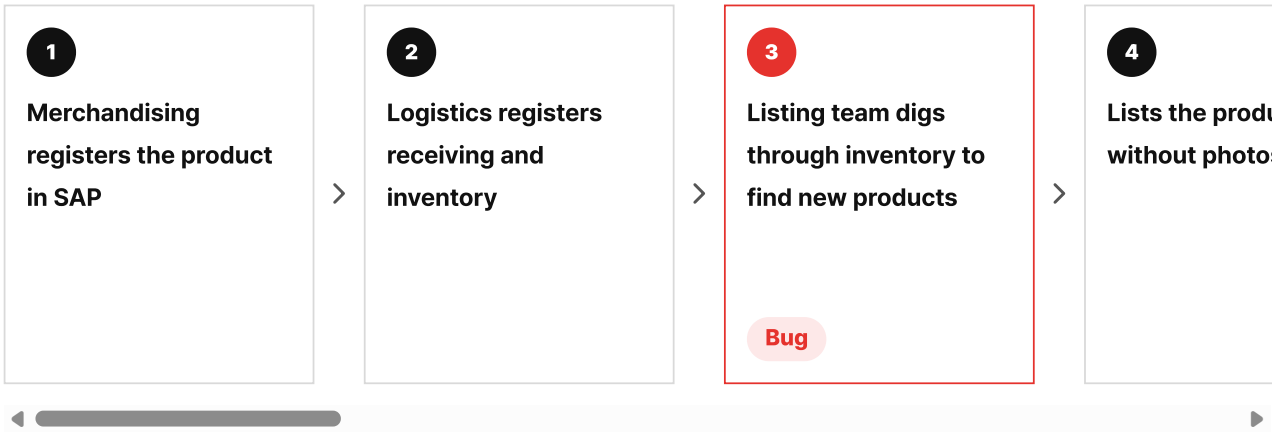


[AS-IS] The old listing process: the same information re-typed by people across five teams. The seven red boxes are the core bugs



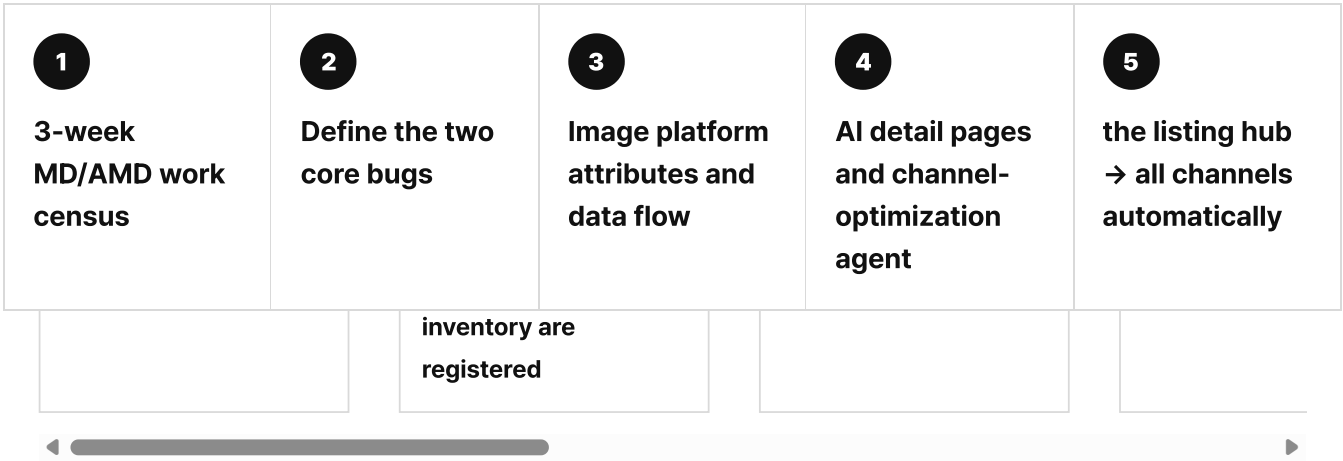
[TO-BE] The the image platform flow after PLM: new-product alerts go out automatically, shots and 3D renders enter through the hub, AI builds the pages, a per-channel agent fits each format and the listing hub sends to every channel. Every red box (core bug) is gone

[AS-IS] The previous business process — seven core bugs



Five teams passed the same information along, re-typing it each time.

Process



Re-typing steps disappear; the listing team only reviews and handles exceptions.

Change ① Lead time from 6 days to 1 hour	Listing lead time from receiving to on sale: average 6 days → at most 1 hour.
Change ② People no longer intervene	Standard listings 100% automated; people review and give AI-logic feedback. Only events and promotions remain, saving 80% of time.
Change ③ Take the photo and AI does the rest	The page is created and listed as the shot is taken; with 3D modeling, products go on sale at planning without a shoot.
Change ④ No more 'received but not on sale'	A monitoring dashboard completely blocks products received but never put on sale.

Change ⑤ Data as an asset, and what it enables

Role & contribution ROLE

All sales-side product data feeds AI training, enabling AI merchandising, recommendations and short-form content, and global channel expansion.

1. Defined what the integrated image platform manages: product code, name, material, color, size, price, inventory and images (model, flat-lay, thumbnail) as the single source of product truth
2. Mapped where data is created and how it flows across SAP, BI, local storage, Google Drive and email
3. Designed the generative-AI workflow: per-item keyword extraction, channel-specific names and promotional copy, image spec adaptation
4. Turned interviews with MD, photo and ops teams into requirements for listing procedures and per-channel rules

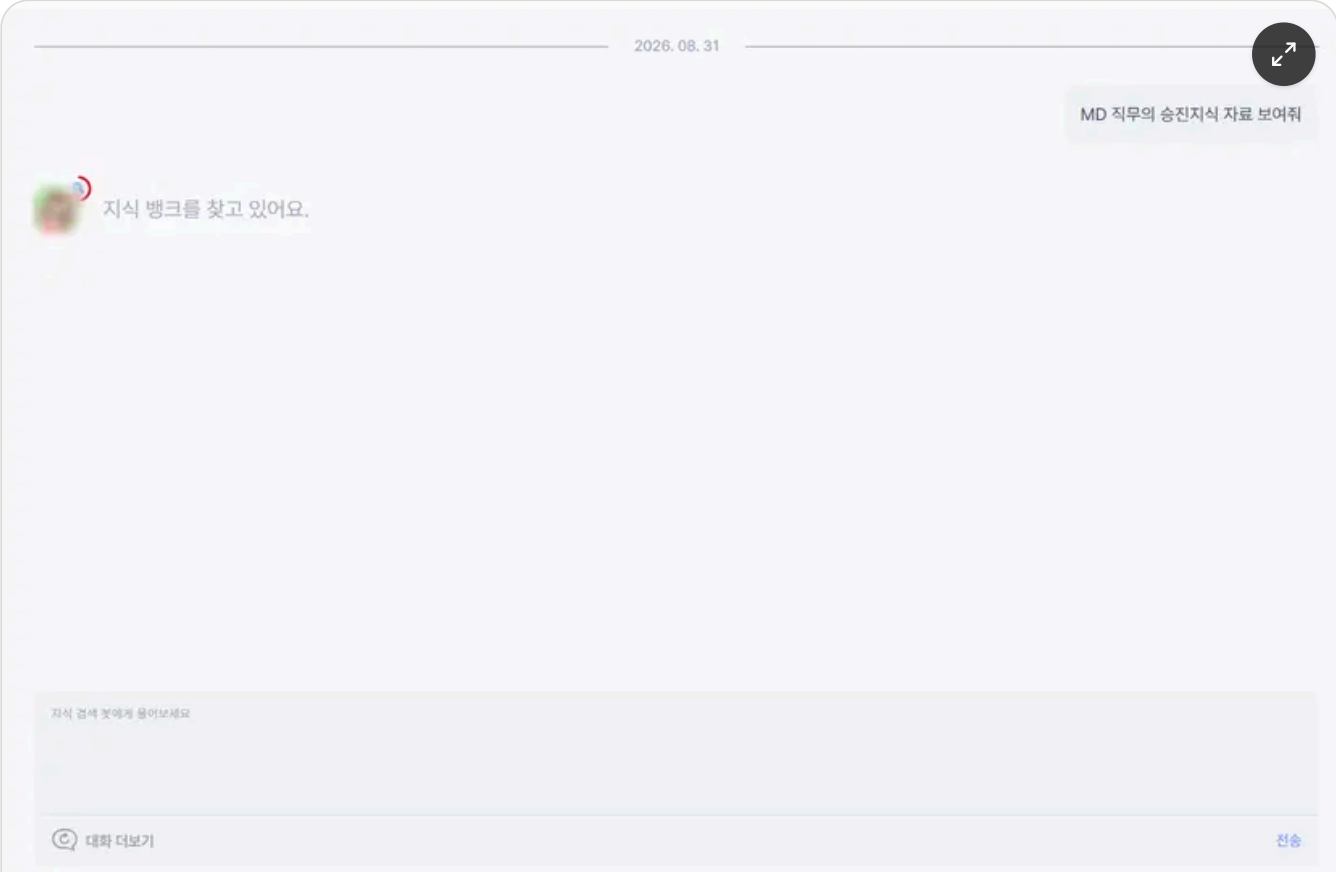
※ All data shown in this portfolio are example figures.

Done · Business Systems

Internal Strategy-Document RAG Search

Not simple search, but a service that searches the organization's decision criteria

1 Problem	2 What I did	3 Result
15 years of strategy decks were image-heavy PPTs that no one could search.	**Built natural-language search on OCR and a vector DB that answers with its sources.**	**90%+ search accuracy, 4.9/5 satisfaction and 98% of the allocated budget saved.**



Finding strategy documents in plain language

Period	2025.01 – 2025.07
Area	RAG · Vector search · Knowledge management
Stack & tools	OCR · Vector DB · GPT API

Purpose PURPOSE

Turned sales data, product information and internal project documents into an AI search and summary service.

Results RESULT

98%

Budget saved

Self-built vector DB vs. allocated budget

90%+

Search accuracy

4.9 / 5

Satisfaction

N=68

7× / week

Usage in core teams

N=68

Background BACKGROUND

An asset without a way to use it: fifteen years of high-quality strategy material had accumulated, but image-heavy PPTs and file-level storage made it hard to search and use.

Project contents CONTENTS

System architecture

Internal documents become JSON, vectorized with the OpenAI Embeddings API into a self-built vector DB. Only chunks close to the question go to the GPT API, which answers with sources.

① Internal source files

Fifteen years of strategy decks, many image-only and unsearchable as is.

PPT · Excel · PDF · Word

OCR / image-to-text

text inside charts and images

② Document processing and JSON conversion

Split into search units and saved as JSON with sources.

Text extraction

Document splitting
chunks per page or slide

Source metadata
file name · page · owning department · date

③ Building the search data

Vectorized with the OpenAI Embeddings API and stored in the self-built vector DB.

Embedding generation
OpenAI Embeddings API

Vector DB storage
self-hosted vector DB (98% below the allocated budget)

④ User question

Plain-language questions, e.g. “find last year's F/W sale strategy deck”.

Natural-language question

User and department
used for permission filters

⑤ Question search

Question vector → similar-chunk search → top results, in in-house search code.

Search code (built in-house)
takes the question, runs the search, organizes results

Embedding model
vectorizes the question the same way as the documents

Vector search
similar chunks from the self-built vector DB, with source and relevance score

Result filtering and ranking
top results only, limited by department, date and permission

⑥ GPT API call

Only retrieved content goes to the model, told not to invent beyond it.

User question + retrieved document content + answer instructions

⑦ Final answer

Shown with source documents and pages for a jump to the original.

Answer

Source documents and pages

How a question is processed

1

Question

Ask in plain language.

2

Question embedding

Vectorized with the same model as the documents.

3

Similar-document search

Closest chunks retrieved from the vector DB.

4

Filtering and ranking

Top results kept, then filtered by permission and more.

5

GPT API call

Question + documents + instructions produce the answer.

6

Answer with sources

Source documents and pages shown below the answer.

Process

1	2	3	4	5
Unstructured knowledge	**OCR / image-to-text**	**Vector DB**	**GPT API natural-language search**	**Iterative user testing**

Role & contribution ROLE

1. Converted strategy documents into JSON
2. Planned the UI/UX of the search screen
3. Iterated on preprocessing and retrieval logic through repeated user tests

※ All data shown in this portfolio are example figures.

Done Strategy Projects

Redefining the Naver Channel Sales Strategy

Why it was customer-structure redesign, not channel expansion

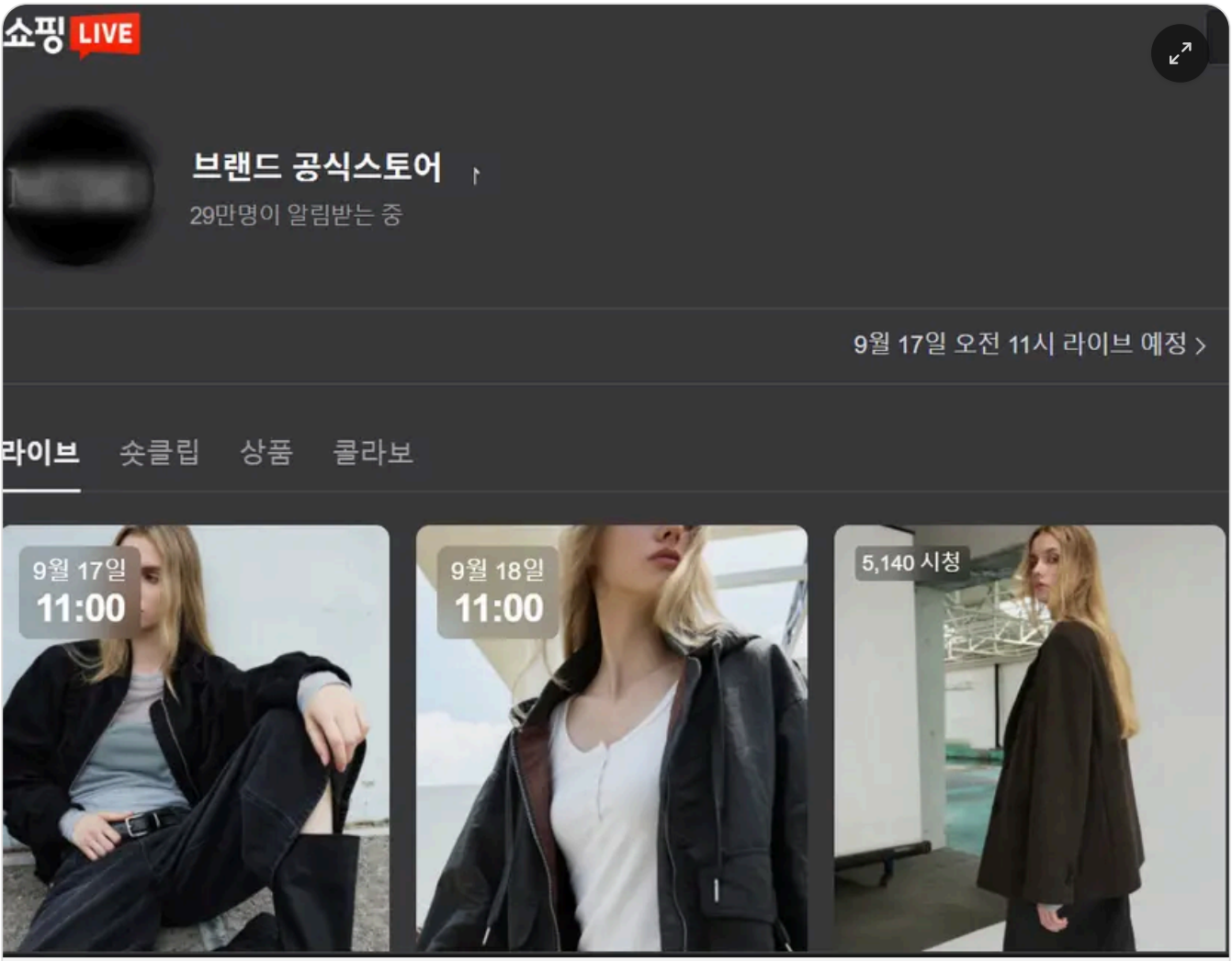
1 Problem	2 What I did	3 Result
Exposure kept growing, but live-commerce conversion stalled at 0.8%.	**Analyzed why customers in their 40s came in; redesigned products, content and pricing.**	**Live conversion rose from 0.8% to 1.9%; sales per show from ₩100M to ₩290M.**

Glossary

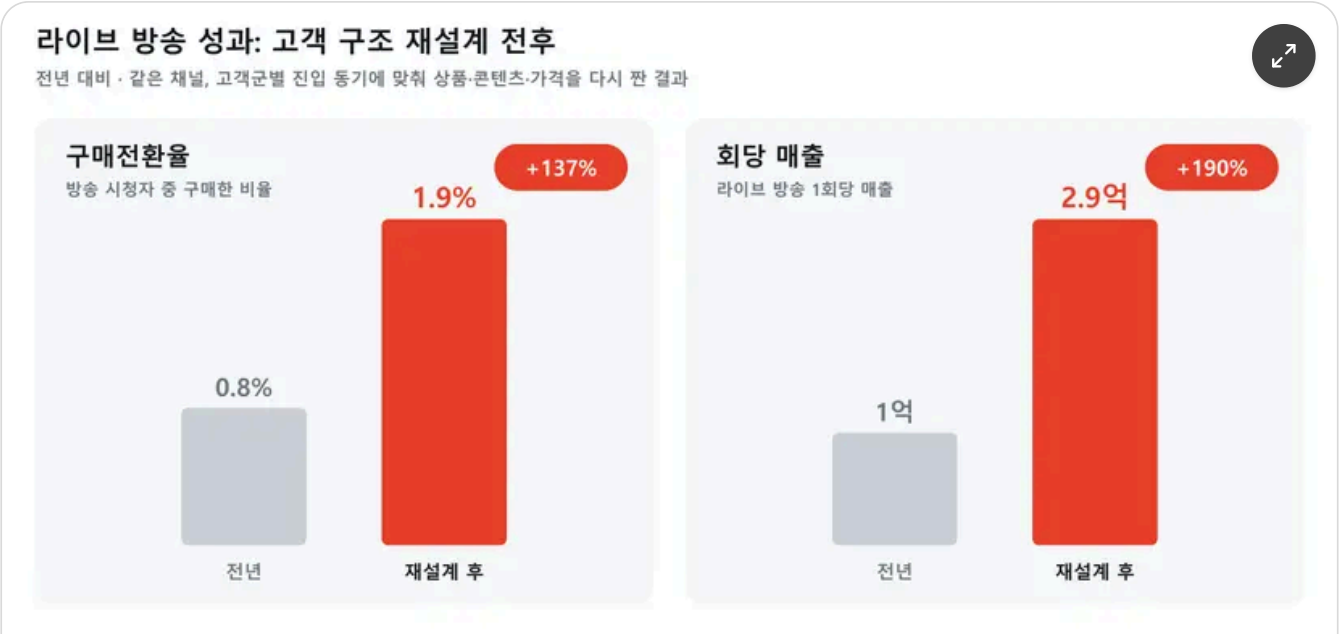
Conversion rate · Share of visitors who actually buy

Live commerce · Selling products through real-time video broadcasts

Entry motive · Why a customer came into the channel



Naver Shopping Live channel



Live commerce results: conversion 0.8% → 1.9%, sales per show 100M → 290M won (year on year)

Period	2024.01 – 2024.02
Area	Customer analysis · Channel strategy · Live commerce

Purpose PURPOSE

Redesigned product, content and channel messaging from an analysis of each customer group's entry motives, lifting channel sales.

Results RESULT

137%

Live-commerce conversion growth

YoY, 0.8% → 1.9%

190%

Live-commerce sales per show

YoY, ₩100M → ₩290M

Background BACKGROUND

Growing online sales called for expanding into new online channels with a strategy to match. Online sales had been concentrated among customers aged 20–35, so a strategy to broaden both online channels and online customers was needed.

Role & contribution ROLE

1. Reinterpreted the limits of platform-led growth as a customer-structure problem
2. Analyzed purchase criteria and entry motives of customers in their 40s
3. Redesigned content messaging, review strategy and pricing around customers' purchase criteria

※ All data shown in this portfolio are example figures.

Done Strategy Projects

the sports brand Padding Rebranding Strategy

Designing core products from big data

1 Problem

Planning repeated last year's best-sellers and missed why young buyers chose jackets.

2 What I did

Read 3 years of sales, review and search data to redefine core products and messaging.

3 Result

Sales among customers under 30 grew 100%.

Glossary

Benchmark · A competitor case studied as a model to learn from

Synthetic fill · Padding made of polyester and other man-made fibers

Off-season padding · Padded jackets bought outside winter

Lightweight jacket · A thin, light padded jacket



Market analysis: shrinking padding demand, growing off-season padding demand

01 [시장] 한국의 경제 성장 '둔화'

한국의 경제 성장률은 2024년 2.5%에서 **2025년 1.9%로 둔화될 것으로 예상**됩니다.
그러나 가처분소득 중 소비지출 비중은 71%로 **생각보다 덜 쓰지도 않는 상황**입니다.

Annex Table 1.1.2. Asian and Pacific Economies: Real GDP, Consumer Prices, and Unemployment
(Annual percent change, unless noted otherwise)

	Real GDP			Consumer Prices[1]		
		Projections			Projections	
	2023	2024	2025	2023	2024	2025
Asia	**5.0**	**4.6**	**4.4**	**2.6**	**2.2**	**2.6**
Advanced Asia	**2.0**	**1.6**	**1.9**	**3.6**	**2.4**	**2.1**
Japan	1.7	0.3	1.1	3.3	2.2	2.0
Korea	1.4	2.5	2.2	3.6	2.5	2.0
Australia	2.0	1.2	2.1	5.6	3.3	3.3
Taiwan Province of China	1.3					
Singapore	1.1					
Hong Kong SAR	3.3					
New Zealand	0.6					
Macao SAR	80.5					
Emerging and Developing Asia	**5.7**					
China	5.2					
India[4]	8.2					
Indonesia	5.0					
Thailand	1.9					
Vietnam	5.0					
Malaysia	3.6					
Philippines	5.5					
Other Emerging and Developing Asia[5]	**4.1**					
Memorandum						
ASEAN-5[6]	**4.0**					
Emerging Asia[7]	**5.8**					

Source: IMF staff estimates.

한국은행 전망 (8월 보고서)

GDP 성장 2024년 **2.4%** → 2025년 **2.1%**

민간 소비 2024년 **2.5%** → 2025년 **2.0%**

→ **11월 28일 1.9%로 더 낮춤**

(단위: 천원, %, %p, 전년동분기대비)

구분		소득 (A)	가계지출	소비지출 (B)	비소비 지출(C)	처분 가능소득 (D=A-C)	증감률	흑자액 (E=D-B)	증감률	흑자율 (E/D ×100)	증감	평균 소비성향 (B/D×100)	증감
2020	1/4	4,366	3,264	2,380	885	3,482	3.7	1,102	35.5	31.7	7.4	68.3	-7.4
	2/4	4,315	3,181	2,385	796	3,519	5.5	1,134	16.0	32.2	2.9	67.8	-2.9
	3/4	4,377	3,284	2,426	858	3,519	4.0	1,093	19.4	31.1	4.0	68.9	-4.0
	4/4	4,362	3,227	2,408	820	3,542	2.3	1,134	13.7	32.0	3.2	68.0	-3.2
2021	1/4	4,384	3,292	2,419	873	3,511	0.8	1,092	-0.9	31.1	-0.5	68.9	0.5
	2/4	4,287	3,308	2,475	833	3,454	-1.9	979	-13.7	28.3	-3.9	71.7	3.9
	3/4	4,729	3,500	2,544									
	4/4	4,642	3,406	2,547									
2022	1/4	4,825	3,496	2,531									
	2/4	4,831	3,508	2,619									
	3/4	4,869	3,721	2,702									
	4/4	4,834	3,625	2,697	928	3,905	3.2	1,209	-2.3	30.9	-1.7	69.1	1.7
2023	1/4	5,054	3,885	2,822	1,063	3,991	3.4	1,169	-12.1	29.3	-5.1	70.7	5.1
	2/4	4,793	3,652	2,691	962	3,831	-2.8	1,141	-13.8	29.8	-3.8	70.2	3.8
	3/4	5,033	3,871	2,808	1,062	3,970	3.1	1,162	1.2	29.3	-0.5	70.7	0.5
	4/4	5,024	3,813	2,833	980	4,044	3.5	1,210	0.1	29.9	-1.0	70.1	1.0
2024	1/4	5,122	3,984	2,908	1,076	4,046	1.4	1,138	-2.6	28.1	-1.2	71.9	1.2
	2/4	4,961	3,811	2,813	997	3,964	3.5	1,151	0.9	29.0	-0.7	71.0	0.7

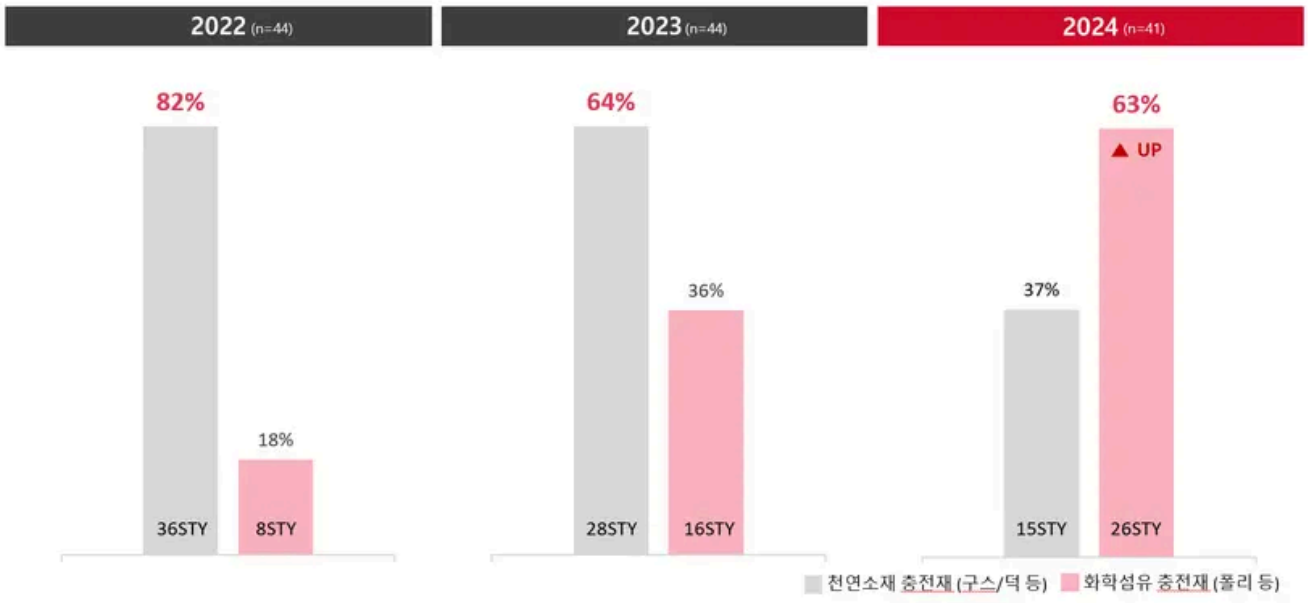
소비지출 비중 71% 한계점

→ **더 쓸 수 없지만, 덜 쓰지도 않는 상황**

[Market] Slowing growth: 1.9% forecast for 2025, consumer spending at a 71% ceiling of disposable income

05 [시장] 매년 늘어나는 '화섬 충전재' 비중

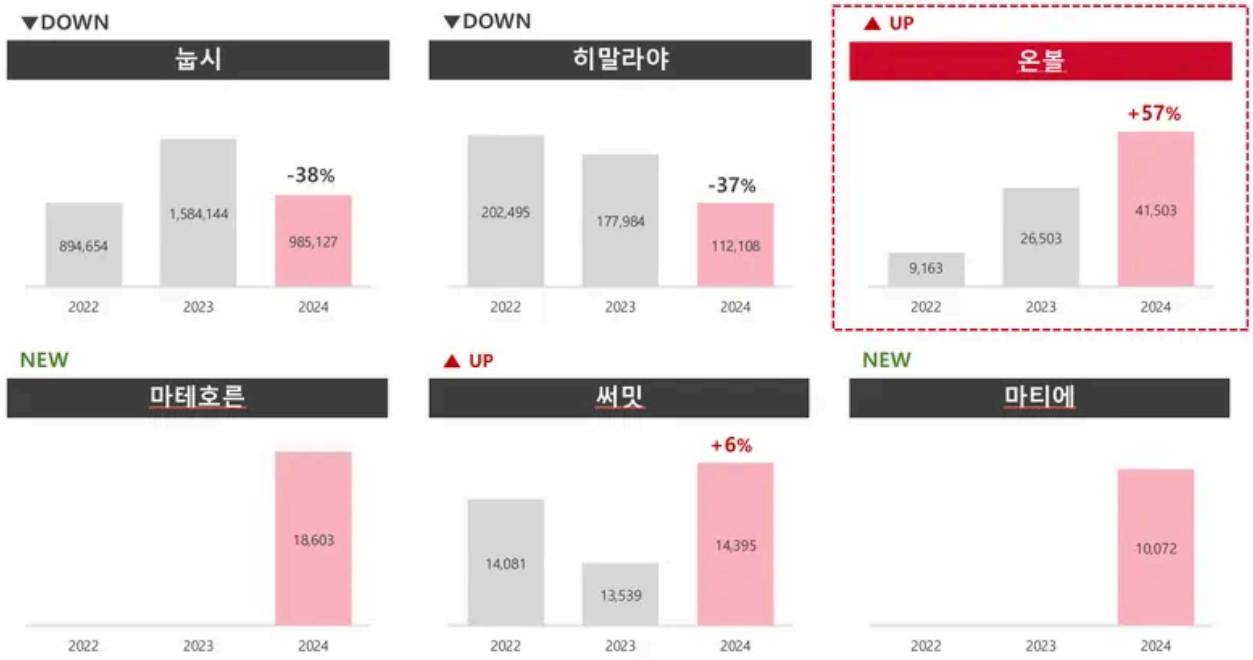
가격 낮아진 이유 뜯어보니, 랭킹 내 원가높은 **구스/덕 천연소재 제품 비중은 매년 줄고** 저렴한 폴리에스터 등 **화학섬유 충전재 비중 높아지고** 있었습니다.



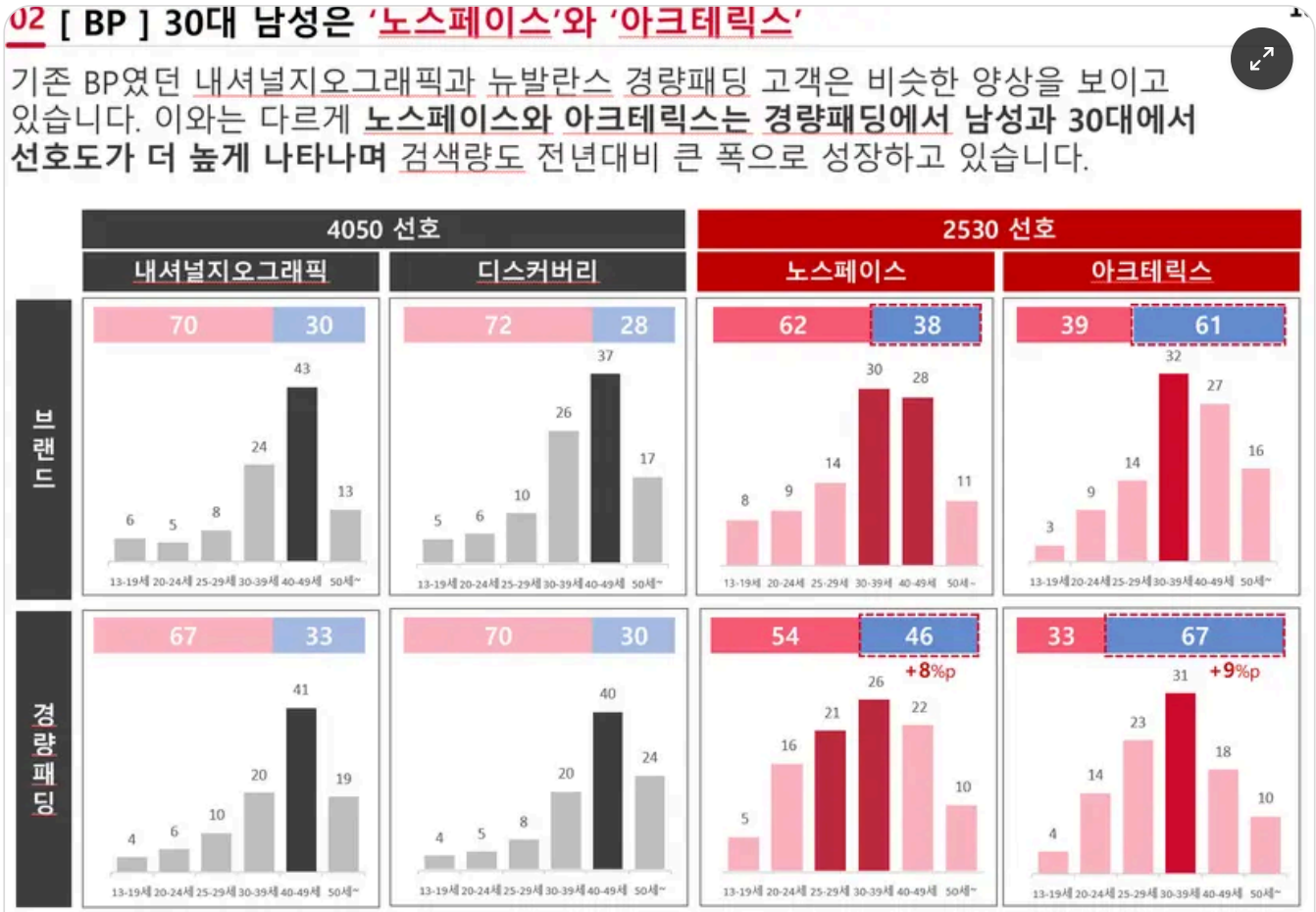
[Market] Synthetic fill rising every year: natural (goose/duck) 82% → 64% → 37%, synthetic 18% → 36% → 63%

08 [BP] 지는 눕시와 뜨는 '온볼'

노스페이스 눕시와 히말라야는 각각 38%, 37% 감소하며 하락세를 보였으나, **온볼은 전년 대비 57% 증가하며 인기를 끌고 있습니다**. 이는 고객들이 패딩 구입을 앞두고 천연충전재인 구스/덕만을 고집하지 않고 **보온성과 가격 모두를 고려**하여 **가성비 높은 인공충전재를 선택**하고 있음을 확인할 수 있습니다.



[Benchmark] Falling Nuptse, rising Onball: down jackets decline while the synthetic-fill Onball grows 57%, a value choice weighing warmth and price together



[Benchmark] Men in their 30s choose The North Face and Arc'teryx: the old benchmarks skew to 40s and women, the new ones to 25–39 and men

Period	2024.11 – 2024.12
Area	Product strategy · Review analysis · Search trends

Purpose PURPOSE

Redefined product strategy and marketing messages from sales, review and platform data by age group.

Results RESULT

100%

Sales growth among customers under 30

25FW vs. 24FW

Background BACKGROUND

Customers wanted better value for the same money, and padded jackets were shifting from natural to synthetic fill.

Our planning carried over last year's top styles, missing customers under 30 and men in their 30s.

Project contents CONTENTS

Four findings

[Market] Growth slows, spending at its ceiling	1.9% growth forecast for 2025, spending at 71% of income. Not closing wallets, but choosing better value for the same money.
[Market] Synthetic fill rising every year	In top-ranked jackets, natural fill fell 82% → 37% and synthetic rose 18% → 63% (2022 → 2024). That is why prices fell.
[Benchmark] Falling Nuptse, rising Onball	The North Face Nuptse -38%, Himalayan -37%, synthetic-fill Onball +57%. A value choice weighing warmth and price.
[Benchmark] Men in their 30s choose The North Face and Arc'teryx	Old benchmarks skew to 40s and 70%+ women, like ours. The North Face and Arc'teryx lightweight jackets: 46% and 67% men, led by 30s.

Process

1	2	3	4	5
Market: economy and spending	**Market: fill materials of top-ranked jackets**	**Benchmark: competitor product trends**	**Benchmark: age and gender preference**	**Redefine core products and message**

Role & contribution ROLE

1. Analyzed men's and women's needs for padding items
2. Analyzed three years of our own padding sales by item
3. Analyzed reviews of top-selling items on the three key platforms

※ All data shown in this portfolio are example figures.

Done Strategy Projects

Customer Churn Analysis for a Premium Brand in China

Six non-customer studies to find ‘brand casual’, localized for white-collar (白领) women

1 Problem	2 What I did	3 Result
The brand kept its formal-heavy lineup, and sales fell 27% every year.	**Ran six customer studies, including non-customer interviews, and localized the products.**	**Sales up 8% year on year, from decline to growth; FW denim order value up 320%.**

Glossary

Non-customer research · Studying people who do not buy the brand

White-collar (白领) · Chinese term for office workers

TPO · Time, place and occasion an outfit is worn for

Brand casual · Premium casual wear fit for both office and weekend

SS · FW · Spring/summer and fall/winter seasons

Shop-along · Shopping with a customer to observe how they buy



Store in China



Key research numbers: formal demand 21% vs 70% supply perception, dual TPO 52%, actual outfit fits

Period	2025.02 – 2025.04
Area	Overseas market · Customer research · Product localization

Purpose PURPOSE

Cross-checked non-customer interviews, street-style surveys, review big data and social media to find why sales fell and whom to win, carried through to SS and FW product input.

Results RESULT

+8%

Sales growth vs. same month last year

from decline to growth

▲320%

FW denim order value

4 styles added

6

Customer-research methods run from zero

46 interviews · 11 shop-alongs · 98 surveys · 177 garments · 2,013 outfits · review big data

Background BACKGROUND

The brand kept decade-old hits such as glossy colored blouses, and sales fell 27% a year.

Office dress had gone casual, but only formal suiting remained: clothes ‘worn one day a week’.

Project contents CONTENTS

Findings and strategy

[Research] Six customer studies + three-year trends	Cross-checked interviews, shop-alongs, surveys, dry-cleaner counts, outfits and review big data. Three-year sales trends, not last season's bestsellers.
[Diagnosis ①] Wearable one day a week, so a waste of money	Office dress went casual; the brand stayed formal. 21% wore formal clothes daily, 70% said it had only formal clothes.

[Diagnosis ②] The customer to win: global-minded white-collar women	Women aged 35–45 at large companies, 44% global-brand buyers. The brand's gold trims and loud colors read as ‘old’.
[PART ① brand casual] Expand to the dual TPO of daily commute and weekend shopping	Dual TPO, where formality and comfort both matter, is largest at 52%, with 69% mid-to-high-priced brand purchases: the opportunity.

[Type and fit] Relaxed fit for formal types, regular fit for the rest

Regular fit reads formal, looser reads casual. Of 2,013 outfits: loose-fit jackets 73%, regular-fit shirts 73%.

Process

1 **Six non-customer studies**	2 **Diagnosis: once-a-week wear · customer mismatch**	3 **PART ① brand casual principles**	4 **PART ② Localizing to white-collar needs**	5 **Basics and differentiators per item**
6 **SS · FW product input**				

baseline; versatility and quality-led luxury decide the purchase. Positioned between SPA and premium brands.

[Execution] SS and FW inputs

Role & contribution ROLE

Proposed cutting formal-led products and rebuilding around shirts, denim and business casual.

1. Defined the cause of declining sales as a shift in customers' workwear norms, not the products
2. Ran six customer-research methods from zero, revising questions and methods each round and wrote them up as the product customer-research validation process
3. Derived steady and trend types per item and customers' top needs and complaints from three years of Tmall top-20 sales data and review big data
4. Documented basics and differentiators for dresses, denim, knits/cardigans, shirts, tees and jackets by learning from benchmarks (Ralph Lauren, Edition, MO&Co., Uniqlo)
5. Decided SS and FW inputs (type, fit, color, fabric) with the local merchandising team

SS: brand casual types added (delivered 31 March). FW: 4 denim styles (order value up 320%), 15 shirt styles, 6 shirt-dress styles.

※ All data shown in this portfolio are example figures.